

TRANSPONDER LANDING SYSTEM (TLS) OPERATORS MANUAL



October 20, 2020

DOCUMENT #020-00072 REV. F

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ENGINEERING CHANGE ORDER RECORD TRANSPONDER LANDING SYSTEM (TLS) OPERATIONS MANUAL

Document #020-00072 Rev. F

Revision	Issue Date	Revision Description	ECN #
A	09/10/12	TLS Manual Division	1916
B	2014Dec09	4.3 Updates	1935
C	2018Apr11	Update customer support phone number	02046
D	2019Oct01	Text changes for clarity, added parallel port switch, combiner, multiple aircraft multiple GTU, GXT4 UPS, new rack diagrams, sensor 920-00400, removed references to RSM and XP, added photos of post January 2017 antenna assemblies	ECA-00148
E	2020Jul8	Update Base 2 OS and Processor	ECA-00177
F	2020Oct20	Update RCU and GTU sections	ECA-00197

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SECTION 1: INTRODUCTION

1.1 Document Overview

This document provides an operational description of the Advance Navigation and Positioning Corporation's (ANPC) Transponder Landing System (TLS) and its operating components. It describes the system, how it performs its functions and how the system is operated.

1.2 ANPC TLS Manuals

The ANPC TLS is accompanied by a group of four primary manuals:

- Document # 020-00071, TLS Calibration Manual
- Document # 020-00072, TLS Operators Manual
- Document # 020-00073, TLS Installation Manual
- Document # 020-00074, TLS Maintenance Manual

The ANPC TTLS includes two additional manuals:

- Document 020-00076 Container System Deployment
- Document 020-00077 TTLS Packing Instructions

SECTION 2: THE TLS

2.1 TLS Overview

The TLS is a ground-based system designed to provide surveillance and precision approach guidance utilizing the aircraft ILS localizer, glide slope, and transponder equipment.

TLS ground equipment consists of a secondary surveillance interrogator, sensor arrays to detect aircraft lateral and vertical position and UHF/VHF transmitters. The equipment is installed near the touch down point for the runway that is served. The footprint can straddle both sides of the runway or be placed entirely on one side. More information about siting TLS can be found in the installation manual.

The TLS detects aircraft within operable range by interrogating Mode-A/C transponders and displays a track of each aircraft position on the RCU. After ATC clears an aircraft for TLS approach, the pilot must tune and identify the TLS localizer frequency that matches the approach clearance. TLS broadcasts ILS-like signals to guide the aircraft along the desired approach path.

The TLS guides up to four aircraft at a time on four ILS frequency pairs. Approach guidance on each TLS frequency pair is valid for one aircraft cleared by ATC for that approach. The concept of operation is very similar to an air traffic controller providing radar vectors and just as with radar vectors, the guidance is valid only for the intended aircraft. The pilot shall tune and identify the frequency that has been authorized. Up to four separate frequency pairs may be configured for one TLS.

The TLS ground equipment tracks aircraft based on their transponder codes and provides localizer course and glide path based on the position of the tracked aircraft. Flying the TLS guidance computed for another aircraft will not provide guidance relative to the approach; therefore, aircrews must not use the TLS signal for navigation unless they have received approach clearance and completed the required transponder code coordination with ATC or as indicated on the approach plate. Navigation fixes based on conventional NAVAIDs or GPS may be provided in the instrument approach procedure for crosscheck.

Any fixed or rotary-wing aircraft with the following equipment can fly a TLS approach:

- an ILS localizer and glide slope receiver,
- a Horizontal Situation Indicator (HSI) or Course Deviation Indicator (CDI),
- a Mode 3/A, compatible transponder.

All TLS glide slope and localizer signal parameters comply with International Civil Aviation Organization (ICAO) Annex 10 standards for Category I ILS.

2.2 TLS Components

The TLS equipment consists of the following elements:

Base Station Shelter	Environmentally controlled shelter housing TLS electronics
Sensors	Two measurement sensors and three antenna locations that receive and measure aircraft transponder signals: • The Elevation Sensor Assembly (ESA) • The Azimuth Sensor Assembly (ASA) with connected Alternate Time of Arrival (ATA) antenna.

Cal/BIT	Calibration/Built-in-Test Transmitter
RCU	Remote Control Unit for entering transponder codes and viewing system status
Uplink	VHF and UHF guidance signal antennas
Interrogator	Interrogation antenna and optional suppression antenna

2.2.1 Maintenance Interface (MI)

The Maintenance Interface (MI) [Figure 2-1] is a program which runs in the Windows operating system. It is used to start the system software, configure and calibrate the system, as well as allow a technician to check the system's health and status and facilitate trouble shooting.

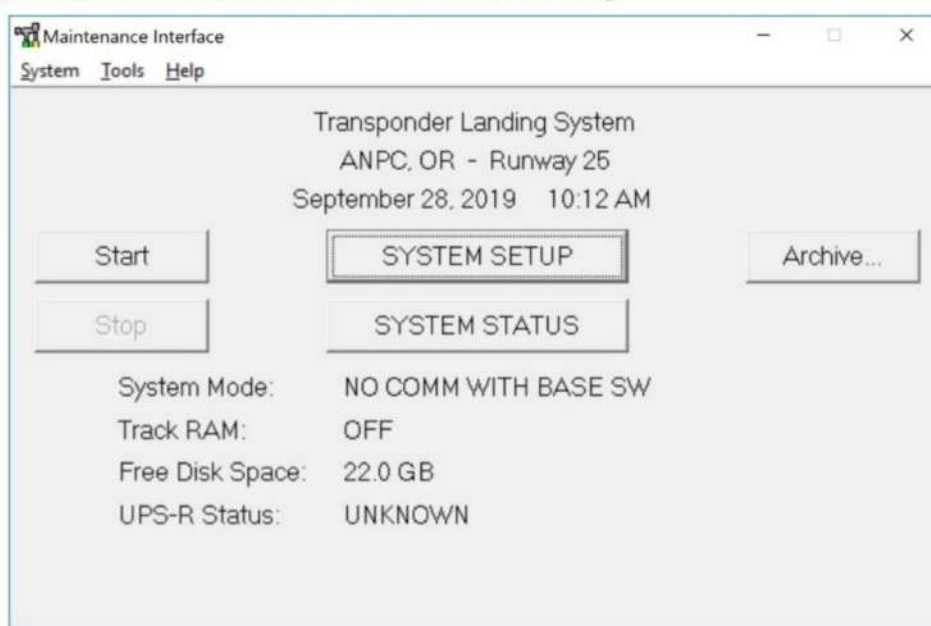


Figure 2-1 Maintenance Interface (MI) Window

2.3 TLS Coverage Volumes

The TLS has two coverage volumes that define how it operates.

1. The surveillance coverage volume (SCV) within which the system tracks aircraft that are within line-of-sight and reply to TLS interrogations. The surveillance track data is displayed on the RCU. Aircraft beyond the SCV are not detected by the TLS. The coverage volume extends to between 60 and 120 miles depending on the version of the TLS sensors installed and type of interrogation antenna.
2. The guidance coverage volume where localizer and glide slope guidance signals can be received by the aircraft. The TLS guidance coverage volume complies with the ICAO standard for ILS. The guidance coverage volume meets and exceeds the standard coverage volumes required by ICAO and extends in some cases much further to meet the requirements for signal coverage for the Instrument Approach Procedure (IAP).

2.4 Electronics racks

Most of TLS system calibration, management, and data processing functions are performed by devices installed in a centralized electronics rack (see Figure 5-1) and stored in a temperature-controlled equipment shelter. Depending on site requirements, a second redundant electronics rack may be installed to reduce system downtime.

2.5 Base Computers

Each TLS electronics rack includes two (2) independent computers, each utilizing a different microprocessor and operating system but running the same TLS software. Each unit provides monitoring, track processing, and guidance control. The outputs are compared in the Guidance Transmitter Unit (GTU) circuitry and the GTU triggers an alarm if there is a difference between the microprocessor outputs. These are not redundant devices but are designed to safeguard against computer-based failures. This design exceeds the design assurance requirements for an ICAO-compliant CAT I ILS.

CPU1 is an Intel™ Pentium processor running the QNX™ operating system. CPU1 also contains the Interrogator Control Card (ICC). CPU2 is a AMD™ processor running the Windows 10 LTSC IoT 2019™ operating system.

The CPUs are the main controllers of the TLS system and perform the following functions:

1. Control the ASA and ESA
2. Control the system timing and Interrogation Transmitter functions through the ICC.
3. Control the GTU(s).
4. Run all tracking, integrity monitor, and interface software.

2.6 Maintenance Interface Unit (MIU)

The TLS electronics rack includes a third central processing unit to function as the Maintenance Interface Unit (MIU). The MIU is used to configure TLS key performance parameters, complete calibration of the GTU's and the system, and to use software tools to help with return to service. When the archive utility is used, the compressed files from Base1, Base2 and each RCU are stored on the MIU.

2.7 Interrogation and Suppression Transmitters and Antennas

The Interrogation Transmitter generates interrogation pulses and side lobe suppression pulses under control from the base CPU1 through the Interrogation Control Card (ICC). The Interrogation transmitter outputs SSR 1030 MHz P1 and P3 interrogation pulses to trigger a reply from each aircraft transponder within operable range. The optional P2 pulse is a 1030 MHz SSR suppression pulse which ensures that only aircraft conducting a TLS approach within the active uplink volume are interrogated by the TLS.

2.7.1 Interrogation/Suppression Antennas

The Interrogation and Suppression (option) Antennas [Figure 2-2] are vertically polarized DME type antennas, mounted on the Interrogation Antenna assembly and connected with flexible coaxial lines from the Interrogation Transmitters installed at the base station. The Interrogation antenna non-suppression option is shown in Figure 2-4.



Figure 2-2 Interrogation and suppression (P2) antennas prior to December 2016



Figure 2-3 Omni interrogation and directional suppression (P2) Antennas after January 2017

2.7.2 Signal Test Points Panel

The Signal Test Points Panel allows for easy signal access from the front of the rack. The panel is used during Factory Acceptance Test (FAT) and during the system life-cycle for troubleshooting as described in the maintenance manual.

2.8 Guidance Transmitter Unit (GTU)

The Guidance Transmitter Unit(s) (GTU) generate the uplink TLS guidance signals to the aircraft with digital control from CPU1 and CPU2. The GTU contains the uplink signal monitor hardware and CPU1 versus CPU2 comparator.

The GTU takes the digital DDM uplink computed by CPU1 and CPU2 and creates the proper balance of 90 and 150 Hz side tones and the Morse code identifier. The GTU synthesizes the VHF and UHF signals and transmits the VHF localizer-like signal on one antenna and similarly the UHF glide slope-like signal on another antenna. This is very different from the traditional ILS that uses a multiple antenna lobe interference pattern to create the traditional ILS signals. The GTU can be programmed to transmit any of the 40 standard ILS signal pairs for localizer and glide slope. The monitor signal is returned to the Base CPU's as part of the integrity monitor. Another part of the built-in integrity monitor is a hardware watchdog timer that shuts off the transmitter outputs if CPU controls to the GTU become inactive. The GTU can be used to generate localizer audio marker tones that the pilot can receive through the localizer audio for range verification.

Each electronics rack can be configured to have up to four GTU's to guide up to four aircraft at a time. For the racks that have multiple GTUs, a combiner is included that combines the signal from multiple GTUs for transmission on a single antenna. For example, each GTU outputs VHF localizer and each of those outputs is combined to transmit on a single localizer antenna. For multiple GTU operation, a parallel port switch is included to multiplex GTU commands from each base computer (CPU1 and CPU2) to each GTU. In this configuration the uplink antennas will have a higher gain in order to compensate for the loss of signal in the combiner.

2.9 Reserve Rack and Rack Switch Control Options

A reserve electronics rack is supplied as an option. For this configuration, there is a Rack Switch Control (RSC) assembly that transfers input/output from the currently selected rack to the alternate. The RCU is used to activate the RSC and switch racks. There is an Ethernet switch dedicated to serving each electronics rack and a third Ethernet switch used for communications with the TLS sensors (ESA and ASA) and the RCU. The rack switch is connected to only this third Ethernet switch such that network communications can be selected for Rack A or B. The RSC has system T-Zero fiber optic, Ethernet communications and RF signals and routes them to the active rack. If both racks remain powered on, then if an electronics failure occurs in the active rack, the RSC can be used to switch the signals to the other rack. One rack may be powered off depending on the customers preference for having a hot spare or just a spare electronics rack.

2.10 Uplink Tower with Localizer and Glide Slope Antennas

Localizer: A horizontally polarized antenna installed twenty four (24) feet above the ground and connected with a flexible coaxial line from the base station transmits the localizer signal.

Glide Slope: A horizontally polarized antenna installed twelve (12) feet above the ground and connected with a flexible coaxial line from the base station transmits the glide slope signal.

An integrated monitor antenna is associated with each transmit antenna. A flexible coaxial cable returns localizer and glide slope signals to the monitor circuits of the GTU in the base station.



Figure 2-4 Uplink Tower: Standard gain Localizer (top) & Glide Slope (bottom) Antennas used with single GTU configurations



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Figure 2-5 Uplink Tower: high-gain Localizer (top) & Glide Slope (bottom) Antennas used with multiple GTU configurations

2.11 Uninterruptible Power Supply

Each rack contains an Uninterruptible Power Supply (UPS). The UPS is a 3 KVA unit and includes a battery reserve that provides more than 60 minutes of system operation if a facility power outage occurs. All TLS equipment, including obstruction lights, is powered through the UPS. With the Reserve Rack option and the RSC installed, there are a total of three UPS units. For a single rack, the Cal/BIT and both the ESA and ASA are powered through a breaker panel by the single UPS. For the dual rack configuration, the Cal/BIT, ESA and ASA are powered by the rack switch UPS.

2.12 Maintenance Interface Unit

The Maintenance Interface Unit (MIU) uses the Windows operating system. The MIU is accessed by the same monitor and keyboard (integrated with the KVM tray) that service CPU1 and CPU2. The MIU provides the maintenance technician with diagnostic tools and system control for use in configuring, troubleshooting and maintaining the TLS.

2.13 Calibration/Built-In-Test Transmitter

The Calibration/Built-In-Test (Cal/BIT) transmitter and antenna provide the 1090 MHz monitor signal that is received by the AOA sensors for built-in-test. The Cal/BIT Transmitter is housed in an environmentally-sealed enclosure. The Cal/BIT antenna is a vertically polarized antenna, and is triggered to transmit via fiber optic line from the base station. The Cal/BIT is light in weight and frangible and may be installed in the runway safety area (RSA) or obstacle free zone (OFZ) near the runway.



Figure 2-6 Cal/BIT Transmitter & Antenna Structure until December 2016



Figure 2-7 TTLS Cal/BIT Transmitter & Antenna Structure after January 2017 – TLS tripod legs include frangible coupling



Figure 2-8 TTLS Cal/BIT Transmitter I/O ports

2.14 Remote Control Unit (RCU)

The Remote Control Unit (RCU) is a computer running the Windows operating system and provides the approach controller interface for TLS. Up to five RCU's may be used with a TLS. Each RCU can control up to four GTU's for simultaneous guidance and multiple RCU's can control each GTU. Each RCU can also provide GCA guidance when not providing ILS guidance. The RCU computer is networked to the TLS base station using wired or wireless Ethernet and can display aircraft positions in the system's service

volume. The RCU should be installed near the approach controller and provides the following controls and indicators:

- Used to start the approach guidance when the approach controller clears an aircraft for TLS approach. The LRU Status display indicating the system's general health to the operator with color coding of LRU's.
- The color-coded transponder code display provides the operator with the transponder codes currently being tracked.
- The rack switching function is controlled through the RCU.
- Audio alerts to warn the operator of a system failure.



Figure 2-9 RCUx Display (Plan View)



Figure 2-10 Remote Control Unit

The assignment of RCU's to GTU frequency can be configured through the MI setting interface dialogue. Each GTU can have up to five RCU's assigned to it. It is possible to guide up to four aircraft simultaneously on four separate ILS frequency pairs.

ILS Settings									
GTU Data									
	Approach Designator	RCU Assignments							
GTU1	TLS VECCU 1 A 00	☒ RCU1	☒ RCU2	☒ RCU3	☒ RCU4	☒ RCU5	Select All	Clear All	
GTU2	TLS VECCU 2 A 00	☒ RCU1	☒ RCU2	☒ RCU3	☒ RCU4	☒ RCU5	Select All	Clear All	
GTU3	TLS VECCU 3 A 00	☒ RCU1	☒ RCU2	☒ RCU3	☒ RCU4	☒ RCU5	Select All	Clear All	
GTU4	TLS VECCU 4 A 00	☒ RCU1	☒ RCU2	☒ RCU3	☒ RCU4	☒ RCU5	Select All	Clear All	

Figure 2-11 ILS settings window

2.15 The AOA Sensors for the ESA and ASA

The AOA sensors record time-of-arrival, amplitude, frequency and differential carrier phase measurements from the Cal/BIT and the aircraft, then transmit this data to the Base CPUs via fiber optic cable. The 920-00256 AOA sensor electronics are installed in an environmentally controlled cabinet.

2.15.1 Azimuth Sensor Assembly

The Azimuth Sensor Assembly (ASA) is comprised of three vertically-polarized antennas and a fourth antenna known as the ATA. The antennas are light in weight and frangible and can be positioned inside of an RSA or OFZ. The ASA receives transponder reply data and transmits that data via fiber optic Ethernet

link to the base station where the TLS software uses this data along with TOA data from the ESA to determine the azimuth (horizontal position) of the aircraft.



Figure 2-12 Azimuth Sensor Assembly



Figure 2-13 Azimuth Sensor Assembly



Figure 2-14 Azimuth Time-of-Arrival Assembly



Figure 2-15 Azimuth Time-of-Arrival Assembly (ATA) being assembled

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2.15.2 Elevation Sensor Assembly

The Elevation Sensor Assembly (ESA) is comprised of four high-gain, vertically-polarized, antennas. The antennas are mounted on a 30 foot tower with options to be free standing or guyed. The ESA receives Cal/BIT and transponder reply data that the TLS uses to determine the elevation of the aircraft.



Figure 2-16 Elevation Sensor Assembly

2.16 General Equipment Specifications

Table 2-1 presents a summary of TLS system signals and key characteristics.

Table 2-1 Reference Data

Item	Signal Parameter
<u>Uninterruptible Power Supply</u>	
Input Voltage	120 VAC $\pm$ 10%
Output Voltage	120 VAC $\pm$ 10%
Power	3 KVA Maximum
System Time on Battery Power	$\geq$ 60 minutes
<u>Interrogation Transmitter</u>	
Frequency	1030 MHz
RF Peak Pulse Power	58 dBm
Pulse Timing	Per ICAO standards and tolerances
Interrogation Rate	22 Hz precision approach 2 Hz surveillance
<u>GTU - Localizer</u>	
Frequency	108.1 - 111.95 MHz
Transmitter Output Power	24 dBm
Modulation (AM)	90/150 Hz ILS Guidance Modulation
<u>GTU - Glide Slope</u>	
Frequency	329.15 - 335.00 MHz
Transmitter Output Power	34 dBm
Modulation (AM)	90/150 Hz ILS Guidance Modulation
<u>Calibration/Built-in-Test Transmitter</u>	
Frequency	1090 MHz
RF Power	40 dBm
<u>Environmental</u>	
Ambient Operating Temperature	
Shelter Equipment	+10° C to +50° C
Outside Equipment	-50°C to +52°C + 18°C Solar Load
Humidity (dew point of 40°C)	
Shelter Equipment	5 – 90%
Outside Equipment	5 – 100%

SECTION 3: SYSTEM USERS

3.1 System Users

For operations the system has four user groups, each with its own duties and all interactively involved. They are the Air Traffic Controller (ATC), the Pilot, the Approach Controller, and the Ground Control Approach controller.

3.1.1 Air Traffic Control

Is a service provided by ground-based air traffic controllers who direct aircraft on the ground and through controlled airspace, and can provide advisory services to aircraft in non-controlled airspace.

3.1.2 Approach Controller

The approach controller operates between the en-route and the immediate airport airspace areas. The approach controller uses the TLS surveillance information to sequence and authorized aircraft to use TLS.

3.1.3 Aircraft Pilot

The pilot operates the aircraft and must tune and identify the TLS frequency that was authorized for use by the approach controller.

3.1.4 Ground Controlled Approach (GCA) Controller

A GCA approach is a type of service provided by air-traffic controllers whereby they guide aircraft to a safe landing, in adverse weather conditions, based on the radar image on the RCU. By following controller commands to keep the landing aircraft on both glidepath and approach centerline, a pilot will arrive precisely over the runway's touchdown zone. In order to ensure continuous radio communication integrity, controllers are required to make radio transmissions at certain minimum intervals depending on the type of approach flow and phase of the approach. Because of their labor-intensive nature—one GCA controller is normally required for each aircraft on final approach. Up to four RCU controller stations are supported with TLS.

SECTION 4: TLS OPERATIONS

4.1 Introduction

4.1.1 TLS: Unaffected by Signal Multipath

For guidance the TLS computes precision track positions and transmits guidance signals that emulate ILS signals that, for a traditional ILS, would exist at the tracked aircraft's current position. The signal from TLS is a homogeneous signal transmitted from a single antenna, which means the signal is not affected by multipath from obstacles combining with the direct transmission of the guidance signals to the aircraft, unlike an ILS.

4.1.2 Increased Tracking Precision

TLS guidance functionality is provided with more precise aircraft tracking. Guidance utilizes higher interrogation rates and uses measurements of the azimuth and elevation angles of the aircraft transponder signals to calculate a guidance track meeting ICAO Annex 10 ILS Category I accuracy requirements.

The localizer and glide slope corrections are computed based on the guidance track to guide the aircraft to the desired course. This guidance information is transmitted throughout the TLS service volume using the Very High Frequency (VHF) and Ultra High Frequency (UHF) signals modulated with 90 and 150 Hz tones. Aircraft equipped with ILS receivers can tune and fly these needle corrections as they would conduct an ILS approach.

4.1.3 Ground Controlled Approach Data

The system is also capable of providing the data necessary for a Ground Controlled Approach (GCA). The horizontal and vertical aircraft position is displayed on the RCU relative to the approach centerline, in order for a Precision Approach Radar (PAR) operator to provide guidance to the aircraft. The positional data displayed on the PAR has the same precision and accuracy quality that the output for the ILS signal is based on.

4.1.4 One Aircraft for each GTU

The TLS guidance signals are dynamic in that they change with time as the tracked aircraft moves. Another aircraft not being tracked that tunes into the TLS localizer and glide slope frequencies does not receive valid guidance. Guidance coordination and control is managed by instituting air traffic control procedures to ensure proper aircraft spacing and one aircraft on TLS approach at a time.

Each GTU guides one and only one aircraft at a time. Each GTU is configured to a different localizer and glide slope frequency pair. The approach controller authorizes an aircraft to conduct a TLS approach according to the TLS approach plate number, where each approach plate shows the frequency to tune.

4.2 Transponder Code Assignment

The operational sequence for providing guidance to a landing aircraft proceeds as follows. Any aircraft operating under ATC Instrument Flight Rules (IFR) procedures is assigned a transponder identification code.

4.2.1 Interrogation/Reply

Operationally, TLS transmits an interrogation signal which, when received by any transponder in the TLS service volume, will trigger a reply. At the same time the interrogation signal is externally broadcast, a "start pulse" signal is sent internally to the two TLS sensors to begin a data collection cycle from the three antenna assemblies ESA, ASA and ATA. For a period following the start pulse, the sensors store the transponder pulse returns from all aircraft in the tracking area along with carrier signal measurements that allow positions to be computed. Figure 4-2, TLS Block Diagram illustrates this cycle.

4.2.2 Surveillance

When aircraft enter the surveillance volume and their identification codes begin to be recorded by the sensors, the TLS computes their positions. Aircraft position is computed and displayed based on the TOA of the code relative to the start pulse, giving range and azimuth to the aircraft. For surveillance tracks, the elevation estimate is based on the Mode C elevation measurement from the replying aircraft. An altimeter at the TLS electronics rack is used to correct for the base station height and display QNH (MSL) altitude on the RCU.

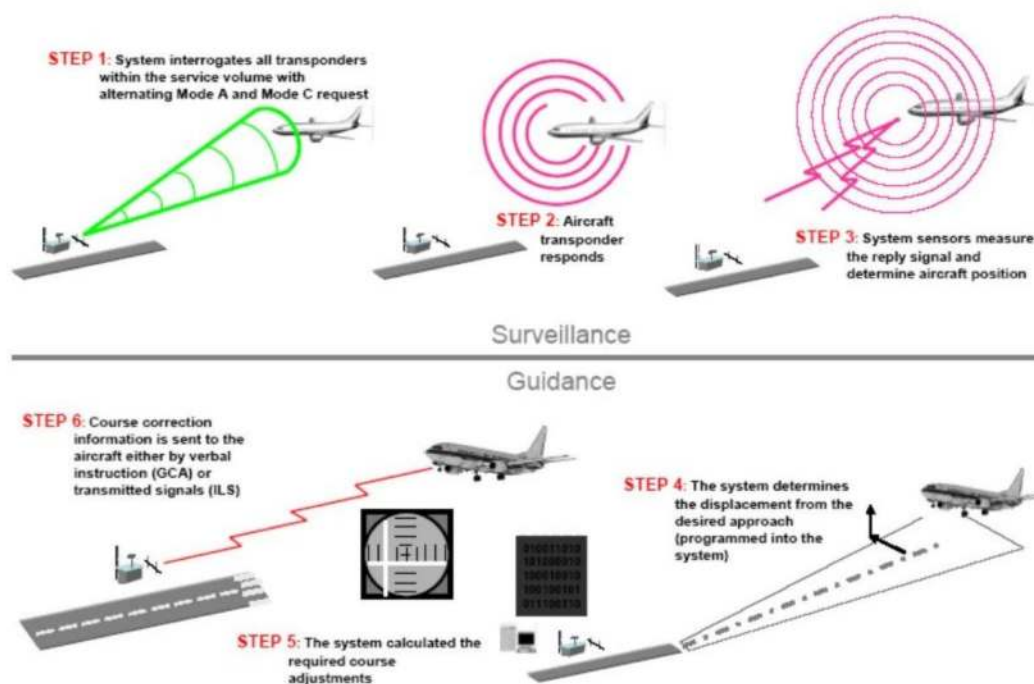


Figure 4-1 TLS operation cycle

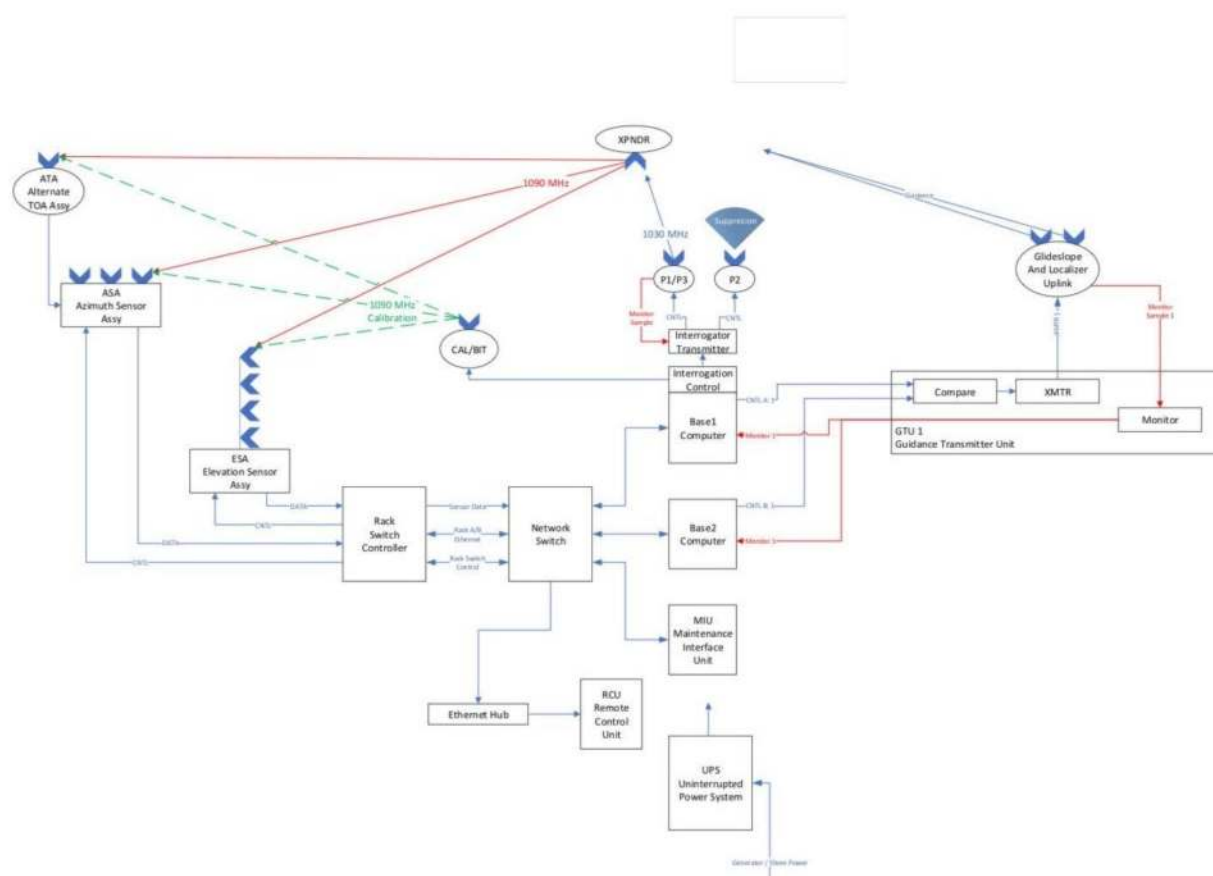


Figure 4-2 TLS Block Diagram

4.2.3 Approach Guidance

When a pilot requests a TLS approach, the approach controller will enter the transponder code in the RCU interface and clear the pilot to conduct the approach. The TLS software searches all of the transponder replies for this transponder code. This interrogation and search cycle is repeated several times each second, the update rate is configurable in the maintenance interface settings menu selection and is normally set to 25 Hz. The transponder reply data is recorded at the sensors and then transmitted to the base station.

Data is then processed using the responding sensor measurements of the aircraft transponder to determine precise azimuth and elevation angles to the aircraft. TLS software then computes the desired needle deflections to guide the pilot toward the desired approach path. The guidance is used to command the GTU signal modulation balance of 90 and 150 Hz tones to emulate an ILS signal that would exist at the tracked aircraft's location. Integrity monitor tests are performed on the sensor data and GTU signal in real time while the guidance transmitter is enabled. Interrogations, position measurements, and guidance transmission updates continue to be generated at the configured update rate. For the pilot, the TLS is flown using ILS techniques. TLS guidance shuts off as the aircraft crosses the runway threshold or when the aircraft leaves the service volume.

It is possible that the integrity monitor would have an alarm condition occur that would require shutting off guidance. For example, if the aircraft transponder does not remain within line-of-sight radar contact, the update rate could drop below a threshold required for precision guidance accuracy to remain within tolerance. In this case, guidance would resume when radar contact is reestablished and the update rate satisfies the minimum threshold.

There is a configurable time-out available on the maintenance interface settings menu that determines the duration of the interval TLS shall look for the transponder code that was input. This time out is normally set to 2 minutes. After the approach controller inputs a transponder code and clears a pilot for landing, the TLS will look for that transponder code for 2 minutes and the aircraft must be within radar contact and within the TLS localizer service volume within 2 minutes or the TLS will transition out of search mode.

4.2.4 Morse and Marker Tones

All radio navigation aids must broadcast Morse station identification and for ILS the Morse must be broadcast on the localizer frequency. Before using the TLS guidance, the pilot must tune and verify that the Morse ident on the TLS localizer frequency matches the published ident on the TLS approach plate. The TLS can also emulate standard ILS marker tones at distances that can be configured in the TLS approach.dat file using the approach design tool. During the approach, the Morse ident is stopped briefly when the aircraft is over a marker in order to broadcast the marker tone. The Morse identification then resumes when the aircraft is no longer over the marker. The Markers are broadcast at a higher volume so that the pilot can turn down the Morse ident and will still be able to hear the marker tone. The TLS marker indications are audio only and do not illuminate the marker lights in the aircraft cockpit.

4.2.5 Pilot Procedures

The TLS approach is flown using ILS techniques. The localizer and glideslope indicators and flight directors react as they would during an ILS approach.

Prior to the Initial Approach Fix (IAF), the pilot should request a TLS approach. ATC will respond with the number of the TLS approach that the pilot shall conduct. The pilot must provide the transponder squawk code and ATC must verify the squawk code. ATC will input the transponder code to the TLS RCU and notify the pilot cleared for the TLS approach that the TLS has been activated. The pilot should check for correct squawk read-back from ATC and ensure that the proper TLS frequency is tuned in the ILS receiver; the pilot must verify the ident of the frequency corresponding to the cleared approach.

Missed approach procedures are flown in accordance with the published procedure and use standard navigation aids other than TLS.

4.2.6 BIT

The TLS has Built-In-Test (BIT) in order to assure that the system is operating properly. During BIT, there is a period of approximately 2 seconds that the TLS Guidance signals are tested. If an aircraft within operable range of TLS is tuned to the TLS guidance frequency during this test, the needles will display a sweeping needle deflection from full scale to full scale for both localizer and glide slope for a few seconds, then return to off (flagged state). In addition, if the pilot is monitoring the audio on the TLS localizer frequency, there could be a one (1)-second long Morse tone heard. This is normal and the pilot can also conclude from this test that the aircraft avionics are functioning properly. The system continues to operate normally if BIT monitors pass their diagnostics.

4.3 TLS Guidance Calculation

The TLS guidance calculation provides an emulation of a standard ILS guidance volume. The approach definition provides the means to achieve course and glide path alignment and displacement sensitivity of the modulated uplink in order to comply with ICAO Annex 10 requirements for an ILS.

4.3.1 Approach Definition

The TLS coordinate system has x aligned with the runway, with the origin at the approach threshold. Since the coordinate system has the same origin and orientation with respect to the runway at every TLS site, approaches may easily be input and verified in advance of their use at a facility. The coordinate system sign convention is defined in Figure 4-3 TLS Approach Coordinate System.

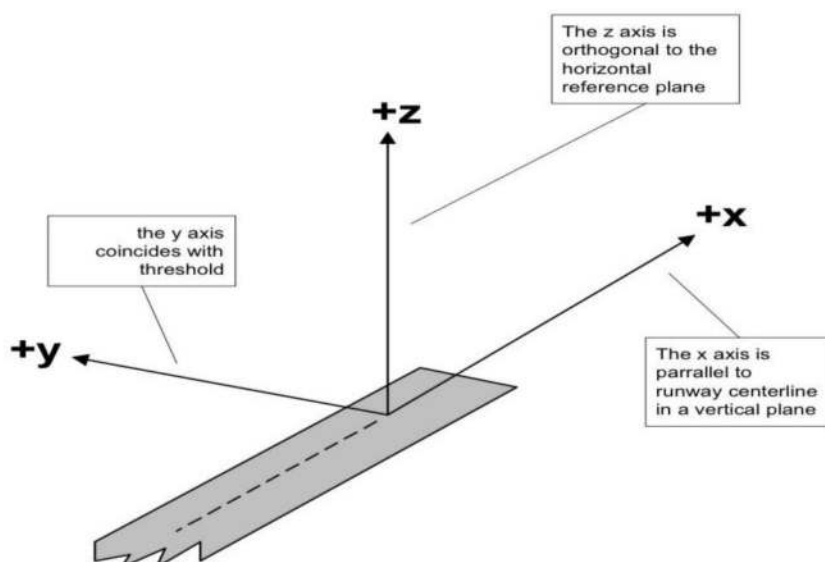


Figure 4-3 TLS Approach Coordinate System

4.3.2 CDI/HSI Deflection Calculation

The approach calculations are determined by a model of the ideal course and path angle using a series of waypoints in the TLS coordinate system. The lines that join the waypoints determine the approach path [Figure 4-4] for the aircraft.

Localizer and glide slope guidance corrections are proportional to lateral and vertical angular deviations, respectively, from the aircraft position relative to a reference surface. For localizer or lateral deviations, the reference surface is a vertical plane containing the approach segment and the apparent localizer position. For glide slope or vertical deviations, the reference surface is a conical surface whose apex is the ground point of intercept (GPI) and whose axis of symmetry is a vertical line passing through the GPI. The loci of points having the same guidance correction are similar surfaces.

The CDI/HSI needle deflections are calculated by computing the angular deviations from the aircraft position to the reference surfaces. The ratio of the lateral angular deviation to the approach course width determines the localizer DDM which should be transmitted to the aircraft. The ratio of the vertical angular deviation to the approach path width determines the glide slope DDM which should be transmitted to the aircraft. These DDM values are output to the TLS uplink transmitter.

The transmitted DDM values are read back by the base station CPU's and monitored to ensure that the course and glide path widths comply with the requirement in ICAO for displacement sensitivity for linearity and alignment.

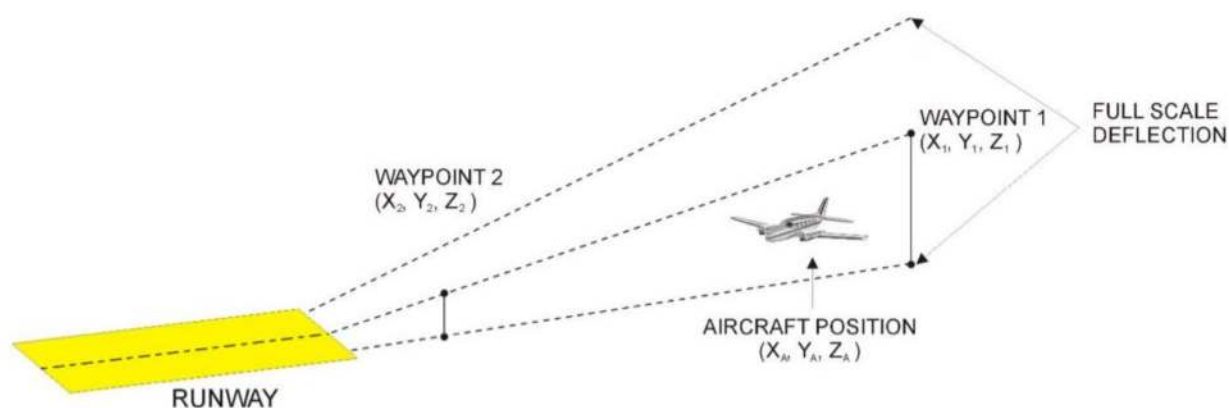


Figure 4-4 TLS Approach Path

4.4 Localizer Transmission

The localizer signal is transmitted to the aircraft corresponding to the most recent guidance correction computed by the system. This signal is equivalent to that which would be produced by a standard VHF ILS localizer, with the exception that these signal properties apply at the position of the tracked aircraft only.

The localizer transmission simulates a composite field pattern which is amplitude modulated by a 90 Hz and a 150 Hz tone. As the runway is viewed from the approach, the 150 Hz tone shall predominate when the tracked aircraft is on the right side and the 90 Hz tone shall predominate when the tracked aircraft is on the left side.

4.4.1 Localizer Frequency

The localizer operates in the band 108.10 MHz to 111.95 MHz. See Table 4-1 for specific frequencies authorized within this band. The frequency variation of the carrier shall not exceed $\pm 0.002\%$, and emissions from the localizer are horizontally polarized.

Table 4-1 Localizer and Glide Slope Frequency Pairings

Localizer	Glide Slope	Localizer	Glide Slope
108.10	334.70	109.50	332.60
108.15	334.55	109.55	332.45
108.30	334.10	109.70	333.20
108.35	333.95	109.75	333.05
108.50	329.90	109.90	333.80
108.55	329.75	109.95	333.65
108.70	330.50	110.10	334.40
108.75	330.35	110.15	334.25
108.90	329.30	110.30	335.00
108.95	329.15	110.35	334.85
109.10	331.40	110.50	329.60
109.15	331.25	110.55	329.45
109.30	332.00	110.70	330.20
109.35	331.85	110.75	330.05

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110.90	330.80	111.50	332.90
110.95	330.65	111.55	332.75
111.10	331.70	111.70	333.50
111.15	331.55	111.75	333.35
111.30	332.30	111.90	331.10
111.35	332.15	111.95	330.95

4.4.2 Localizer Coverage Requirement

The localizer signal covers the azimuth and elevation volumes as presented in Figure 4-5 and Figure 4-6 respectively. The minimum field strength (carrier power) on the glide path and within the localizer course sector, from a distance of 18.5 km (10 NM) to a height of 60 m. (200 ft.) above the horizontal plane containing the threshold, is not less than $90 \mu\text{V/m}$ (-107 dBW/m^2). Over the remainder of the coverage volume, as specified above, the field strength is not less than $40 \mu\text{V/m}$ (-114 dBW/m^2).

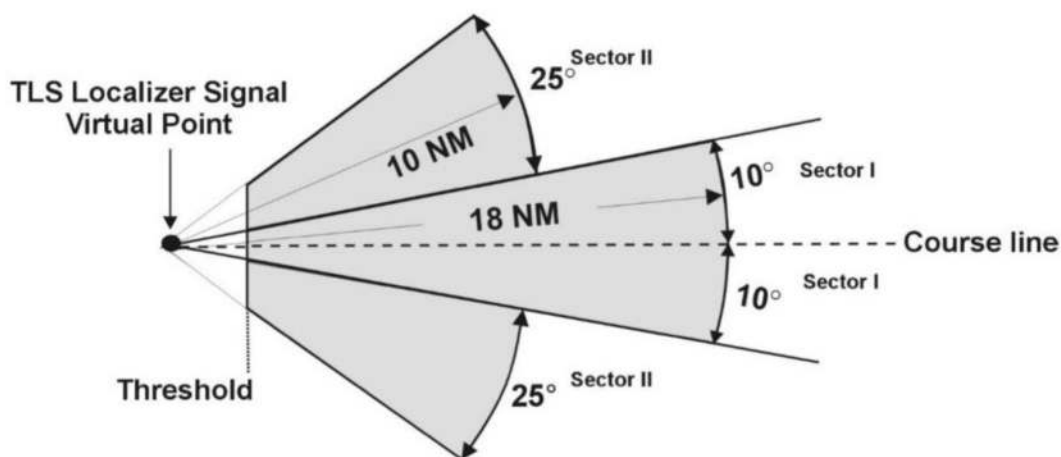
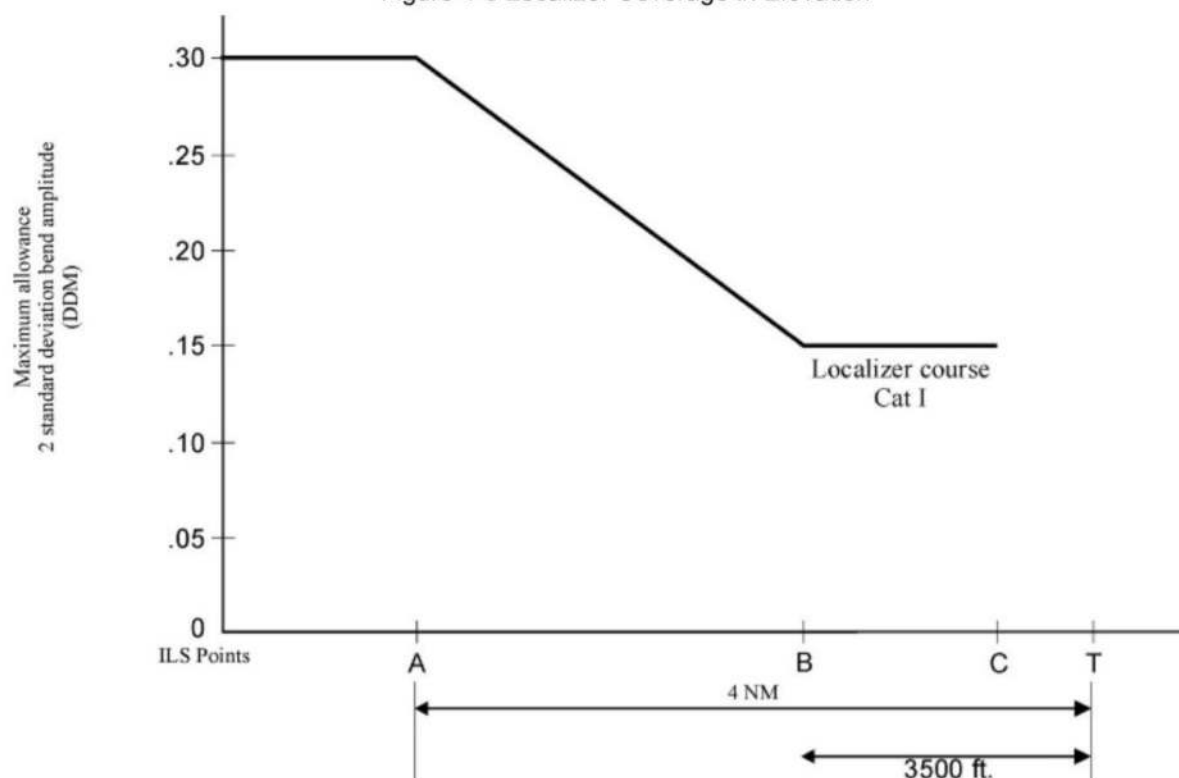
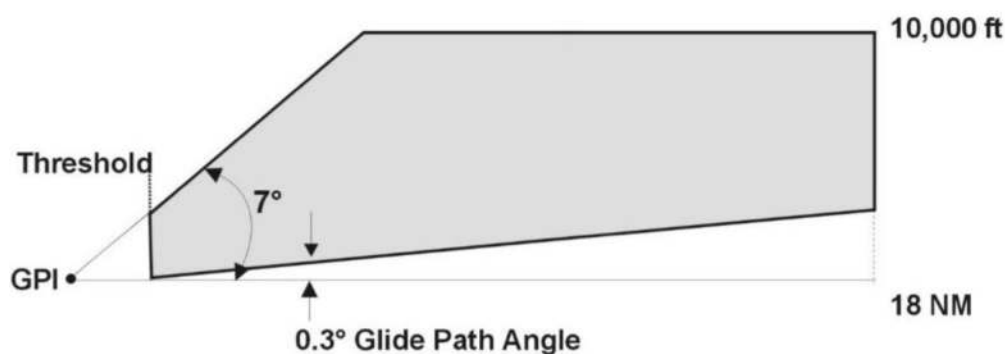


Figure 4-5 Localizer Coverage in Azimuth

4.4.3 Localizer Course Structure

Structure in the course line does not have amplitudes which exceed the following: (Figure 4-7 is DDM, 95% probability).

- Outer limit of coverage to ILS point A -- 0.031 DDM (0.6 deg)
- Decreasing at a linear rate from 0.031 DDM (0.6 deg) at point A to 0.015 DDM (0.3 deg) at ILS point B.
- ILS point B to ILS point C -- 0.015 DDM (0.3 deg)



4.4.4 Localizer Carrier Modulation

The nominal depth of modulation of the carrier due to each of the 90 Hz and 150 Hz tones is 20% along the course line, and depth of carrier modulation for each of the 90 Hz and 150 Hz tones is within the limits of 18 and 22%. The frequencies of the modulating tones are to be 90 Hz and 150 Hz $\pm 2.5\%$, with a total harmonic content of the 90 Hz and 150 Hz tones not exceeding 10%. The modulation tones are to be phase-locked so that within the half sector, the demodulated 90 Hz and 150 Hz waveforms pass through zero in the same direction within 20° of phase relative to the 150 Hz component every half cycle of the combined 90 Hz and 150 Hz waveform.

4.4.5 Localizer Course Alignment Accuracy

The mean course line does not deviate from the runway center line by more than 0.3° (0.015 DDM).

4.4.6 Localizer Displacement Sensitivity

The nominal displacement sensitivity corresponds to a DDM of 0.155 at angular displacements of $3.0^\circ \pm 0.5^\circ$ on both sides of the center course line. The increase of DDM is linear with respect to angular displacement from the course line (where DDM is zero) to an angle on either side of the course line where the DDM is 0.200. From that angle to 35° , the DDM is not less than 0.180. [Figure 4-8].

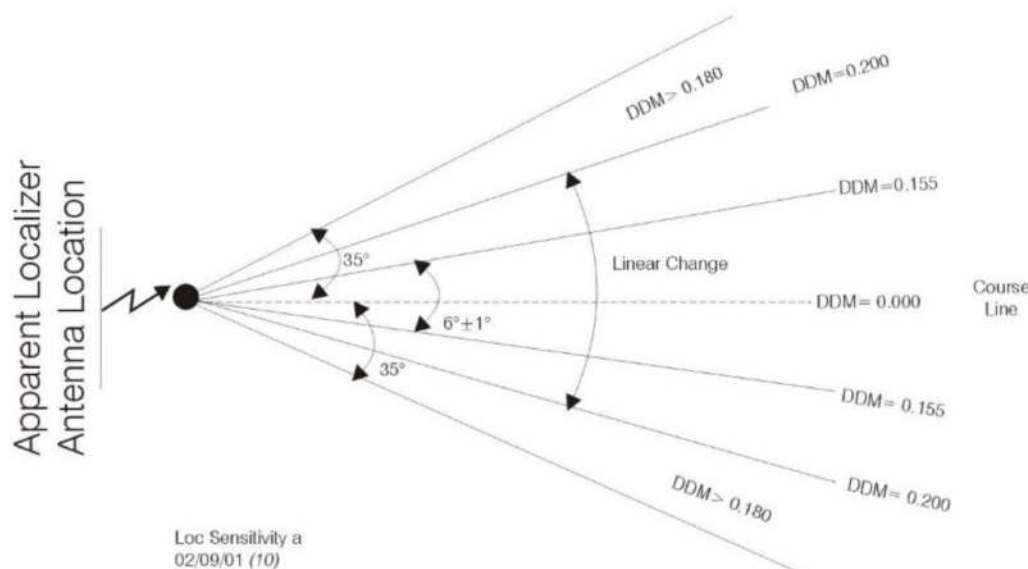


Figure 4-8 DDM and Displacement Sensitivity

4.4.7 Localizer Identification Tone

The identification signal is produced by Class A2A modulation of the localizer carrier using a modulation tone of $1020 \text{ Hz} \pm 50 \text{ Hz}$. The depth of modulation is between the limits of 5 and 15%. The identification signal is International Morse Code and consists of four letters, as assigned by the site controlling agency.

The identification signal is transmitted at a speed corresponding to approximately seven words per minute, and is repeated at equal intervals, not less than six (6) times per minute, at all times during which the localizer is available for operational use. When the transmissions of the localizer are not available for operational use, as, for example, during maintenance, the identification signal is suppressed.

4.4.8 Localizer Audio Indicators

The TLS is capable of adding a single additional modulation tone on the localizer signal which may be used to provide distance or procedure information to the aircraft on approach. This additional tone can be one of the following:

1. The localizer Audio Indicator Tone A signal is produced by Class A2A modulation of the localizer carrier using a modulation tone of $700 \text{ Hz} \pm 50 \text{ Hz}$.
2. The localizer Audio Indicator Tone B signal is produced by Class A2A modulation of the localizer carrier using a modulation tone of $2000 \text{ Hz} \pm 50 \text{ Hz}$.
3. The localizer Audio Indicator Tone C is produced by Class A2A modulation of the localizer carrier using a modulation tone of $2500 \text{ Hz} \pm 50 \text{ Hz}$.

The depth of modulation is between the limits of 25 and 35% for all three tones, and the localizer Audio Indicators are further modulated with a 2-character International Morse Code ID as set via the user interface.

This identification signal is transmitted at a speed corresponding to approximately seven words per minute, and is repeated at equal intervals, not less than six times per minute.

This localizer Audio Indicator will not activate standard marker lights.

4.5 Glide Slope Transmission

The glide slope signal is transmitted to the landing aircraft corresponding to the most recent guidance correction computed by the system. The glide slope signal is equivalent to that which would be produced by a standard UHF ILS glide path, with the exception that these signal properties apply only at the position of the tracked aircraft.

In the following requirements, θ denotes the nominal glide path angle, which is the angle between a straight line that represents the mean of the TLS glide path and the horizontal.

The glide slope transmission simulates, at the aircraft position, a composite field pattern which is amplitude modulated by a 90 Hz and a 150 Hz tone. The reference center of this pattern is a straight-line descent path in the vertical plane containing the center line of the runway, with the 150 Hz tone predominating when the tracked aircraft is below the path and the 90 Hz tone predominating when the tracked aircraft is above the path.

4.5.1 Glide Slope Angle

The TLS employs software adjustment via site-specific configuration parameters to set the glide path angle to match the approach procedure. Nominal glide paths are 2° to 4° with respect to the horizontal, but the TLS can be configured to support steeper glide paths if required by obstructions. Once adjusted, system design ensures the glide path angle remains within 7.5% from the commissioned angle.

Glide Slope Frequency

The glide slope transmission operates in the band 329.15 MHz to 335.00 MHz. See Table 4-1 above for specific frequencies authorized within this band. The frequency variation of the carrier shall not exceed $\pm 0.002\%$, and the emission from the glide path equipment is horizontally polarized.

4.5.2 Glide Slope Coverage Requirement

The glide slope signal covers the azimuth and elevation volumes as presented in Figure 4-9 and Figure 4-10 respectively. The minimum field strength (carrier power) within the above sectors is $400 \mu\text{V/m}$ (-95 dBW/m^2).

4.5.3 Glide Slope Structure

Structure in the glide path has amplitudes no greater than 0.035 DDM (0.14 deg), 95% probability, from the outer limit of coverage to ILS point C.

4.5.4 Glide Slope Carrier Modulation

The nominal depth of modulation of the carrier due to each of the 90 Hz and 150 Hz tones is 40% along the glide path, and does not deviate outside the limits of 37.5% to 42.5%. The modulating tones are 90 Hz and 150 Hz $\pm 2.5\%$, with a total harmonic content of the 90 Hz and 150 Hz tones not exceeding 10%. The modulation tones are to be phase-locked so that within the half glide path sector the demodulated 90 Hz and 150 Hz waveforms pass through zero in the same direction within 20° of phase relative to the 150 Hz component, every half cycle of the combined 90 Hz and 150 Hz waveform.

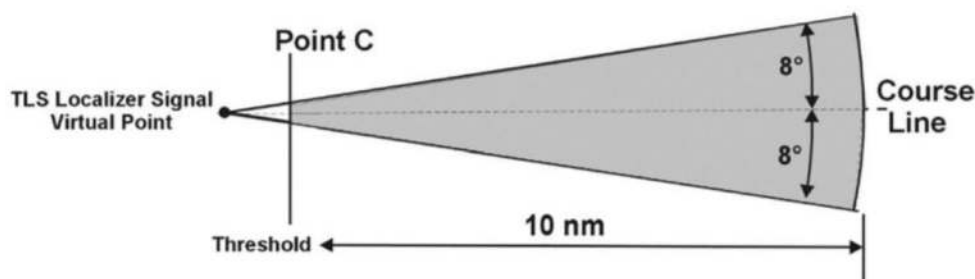


Figure 4-9 Glide Slope Coverage in Azimuth

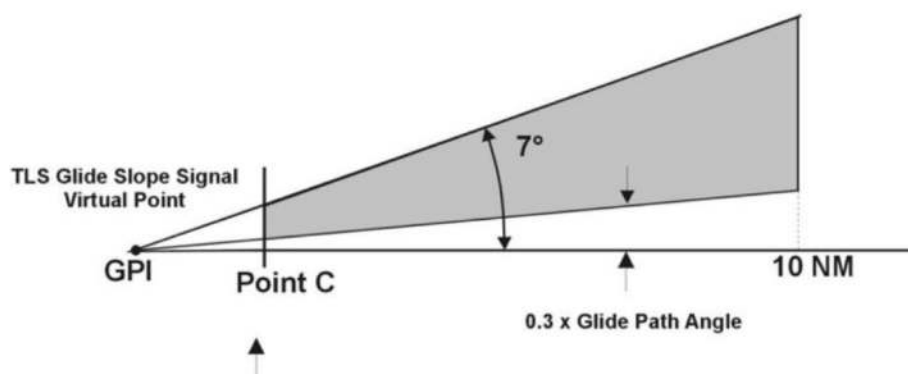


Figure 4-10 Glide Slope Coverage in Elevation

4.5.5 Glide Slope Displacement Sensitivity

The TLS glide slope displacement sensitivity has a nominal displacement sensitivity corresponding to a DDM of 0.0875 at angular displacements of $0.35^\circ \pm 0.07^\circ$ above and below the glide path. The increase of DDM is linear with respect to angular displacement from the glide path (where DDM is zero) to an angle on either side of the glide path where the DDM is 0.250. From that angle to edges of the glide slope coverage volume, the DDM is not less than 0.220.

Beyond distances equivalent to $15 \text{ m}/\tan\theta$, the requirements for precision sensitivity control of the glide slope signal requirements still apply. However, within distances equivalent to $15 \text{ m}/\tan\theta$, the increase of DDM of the glide slope signal shall be linear with respect to distance from the glide path (where DDM is zero) to a distance on either side of the glide path where the DDM is 0.175. See Figure 4-11 for a depiction of the displacement sensitivity.

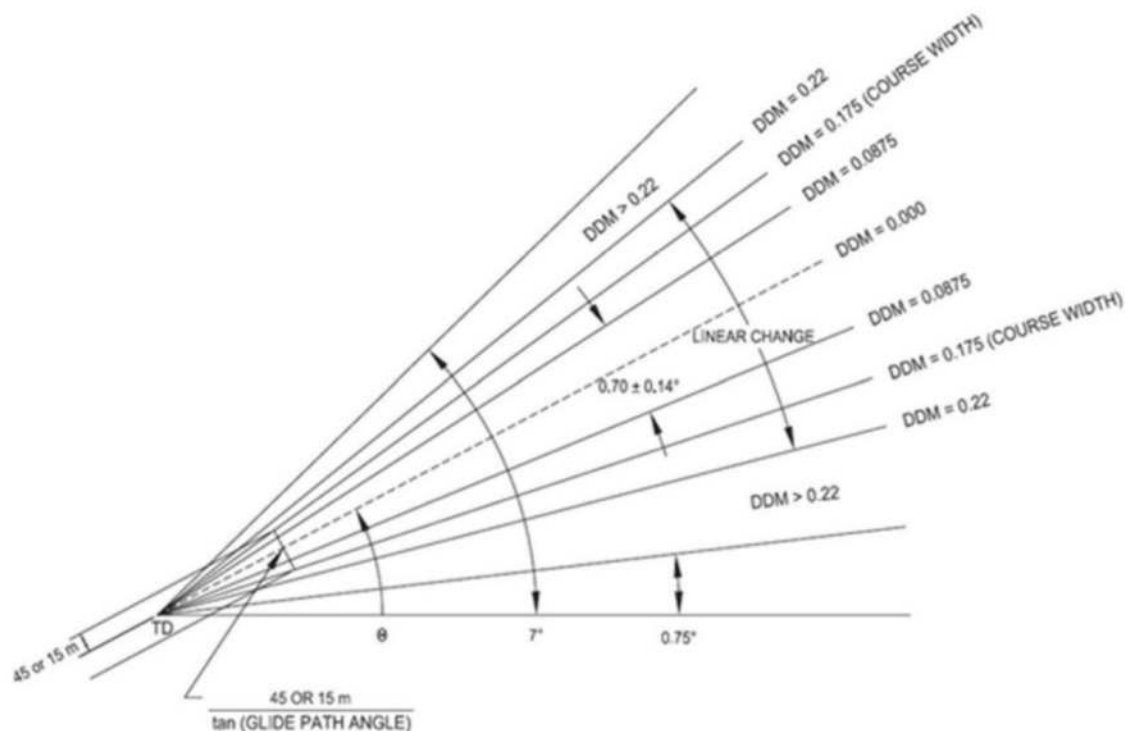


Figure 4-11 Glide Path DDM and Displacement Sensitivity (2)

4.6 TLS Software

The TLS software is comprised of five Software Configuration Items (SCI) associated with the principal TLS components. These SCIs are hosted on different types of processors utilizing different operating systems. Table 4-2 lists the SCIs, their software design assurance levels, Operating Systems (OS), and the processors on which they run.

Table 4-2 Software Configuration Items

SCI	DO-178B Level	Operating System	Processor
Base1	B	QNX™	Pentium™
Base2	D	Windows 10 LTSC IoT 2019™	AMD™
AOA Sensor	D	Texas Instruments RTOS	TMS320C6424
RCU	B	Windows 10 LTSC IoT 2019™	Pentium™
MIU	D	Windows 10 LTSC IoT 2019™	AMD™

4.6.1 TLS Software Environment

Figure 4-12, TLS Software Environment shows the principal elements of the environment in which these SCIs operate. Central control of the TLS system is provided by two Base Station processors (CPU1 and CPU2) and their SCIs. These are dissimilar microprocessors (Pentium and AMD) and operating systems (QNX™ and Windows 10 LTSC IoT 2019™) running programs that contain identical system tracking and guidance computation software.

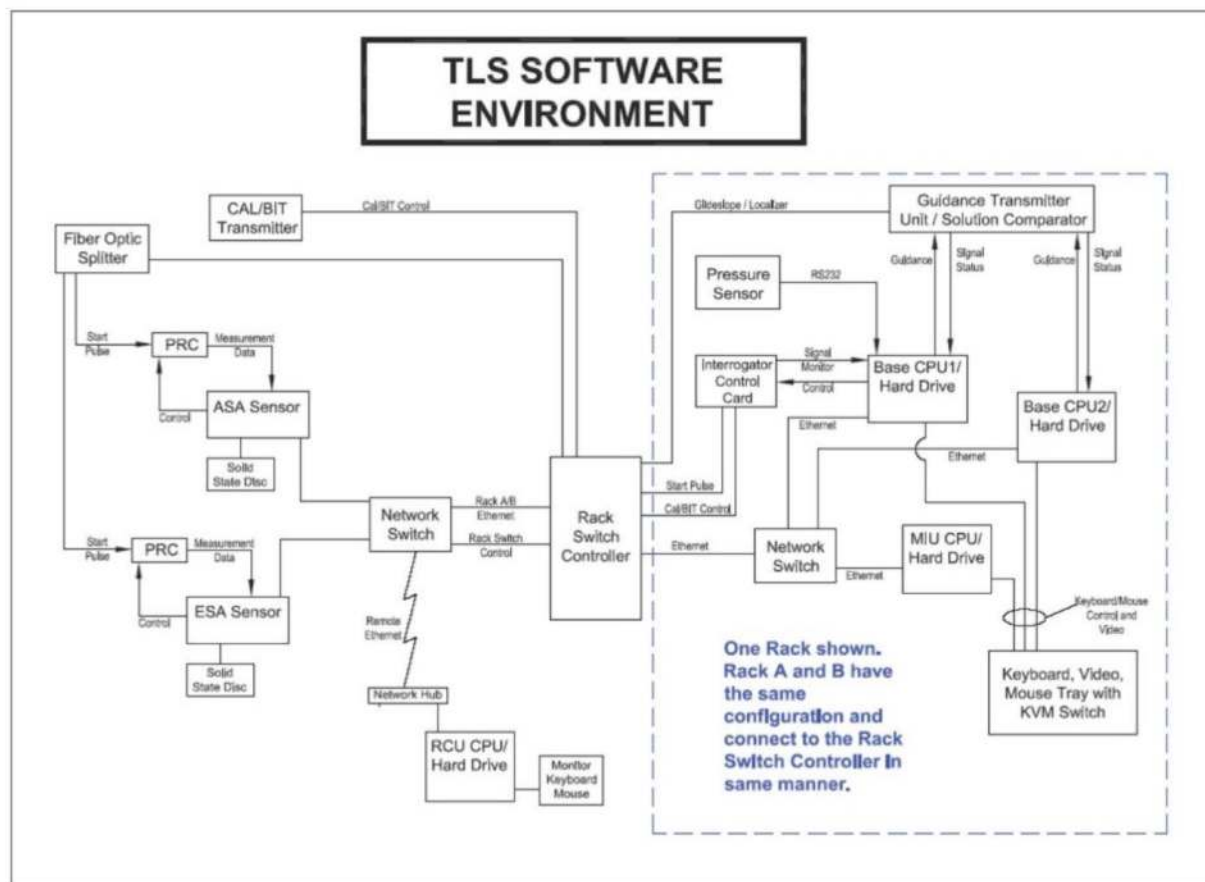


Figure 4-12 TLS Software Environment

4.6.2 Guidance Safety: Two Solutions

For guidance, the two solutions are input to a circuit in the Guidance Transmitter that compares the solutions and stops the uplink to the aircraft if they do not match within a preset tolerance. This is also true for the precision information provided to the RCU. Both solutions are provided, and if they don't match the displayed guidance track will be discontinued.

This design addresses the random hardware or OS failure modes and ensures that guidance transmission will be removed in the event of a failure. The Base Station SCIs provide two parallel computational paths operating on the same sensor data, and producing guidance command outputs that, in the absence of a failure, should be identical.

4.6.3 Sensor SCI

The Sensor SCI is identical for the two sensors (ASA and ESA), which share all functions. The Sensor SCIs control and collect signal measurement data from transponder receivers and antenna arrays and pass it to the two Base Station SCIs via Ethernet over fiber optic cables.

The installation manual has instructions for configuring the software on the sensor to function as an ESA or an ASA.

4.6.4 RCU SCI

The RCU SCI provides the display software portion of the TLS operator functions for normal flight operations. It provides a plan view (top down) display of the terminal area showing tracked aircraft under

surveillance. It provides a means for the TLS Operator to input the transponder code for the aircraft to be tracked, and to initiate and terminate TLS guidance. The RCU display also provides PAR emulation, where the aircraft tracks are displayed relative to the approach and glide path. This display can be used to provide a Ground Control Approach (GCA) to an approaching aircraft that does not have ILS receiver equipment. The display can also be used for man-in-the-loop monitoring of the pilot's actions during an approach and allow the approach controller to provide an extra level of safety oversight of the approach.

The RCU also displays various information of interest to the operator, including TLS operational modes and health status. RCU software allows the operator to automatically switch from a primary TLS Base rack to a backup.

4.6.5 MIU SCI

The MIU SCI provides the software portion of the TLS technician interface to the system for configuring and calibrating the system when first installed and subsequently diagnosing faults and maintaining the system. The Maintenance Interface provides functions for system calibration, real-time monitoring, and troubleshooting.

4.7 System Modes of Operation

The TLS system has several distinct operational modes from which it performs its functions. It is important to understand the distinction between the modes of operation and how they interact. There are five system modes of operation that will be discussed in this manual:

- Initialization
- Diagnostic,
- Maintenance,
- Track, and
- Test Mode.

Guidance is an operational function that occurs during Track Mode.

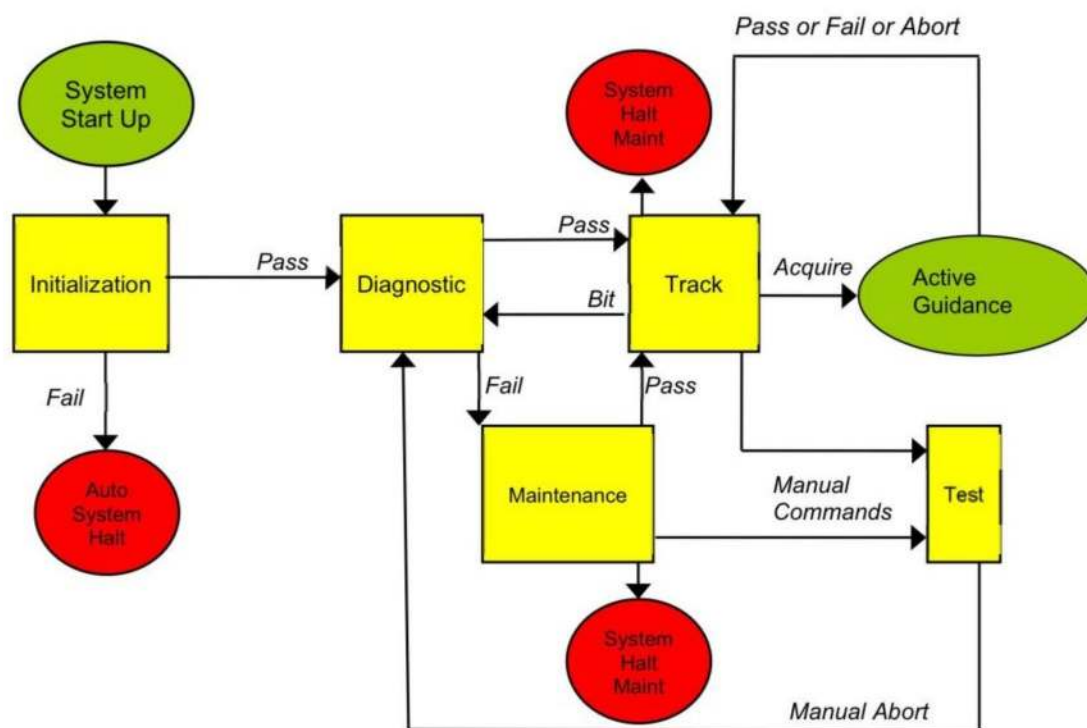


Figure 4-13 TLS Modes of Operation

4.7.1 Initialization Mode

Initialization Mode is automatically entered whenever the system software is started from the MIU. During this mode, the TLS Base Station software performs various start-up procedures, reads and verifies site configuration files, verifies software and configuration file versions, and establishes all network connections. Upon successful completion of all startup functions, the system transitions to Diagnostic Mode. If any of the initialization steps fails, the TLS Base Station software will automatically halt, and the TLS will be unavailable until the failure is fixed.

4.7.2 Diagnostic Mode

After Initialization, the system enters Diagnostic Mode automatically to perform BIT. Successful completion of BIT verifies that the system is ready to transition to Track Mode. BIT may be run for a configurable number of times during a retry schedule if it fails. Failure to successfully complete BIT indicates an operational fault such that the system cannot safely provide the functions it was designed for such as Surveillance and Category I approach. On a failed BIT the system transitions to Maintenance Mode.

4.7.3 Maintenance Mode

The system enters Maintenance Mode only from Diagnostic Mode. This occurs automatically when BIT fails in Diagnostic Mode. BIT continues to run in Maintenance Mode at the same schedule as in Diagnostic Mode. This mode indicates that a system fault exists and that the TLS Maintenance Technician needs to perform TLS corrective maintenance. Transition from Maintenance Mode only occurs following successful completion of BIT and results in the system returning to Track Mode providing surveillance tracking and ready to provide guidance. A manual command from the TLS Maintenance Technician (on the MIU) can cause the system to halt from Maintenance Mode. The Maintenance Technician may also transition the system from Maintenance to Test mode using a manual command from the MIU.

4.7.4 Track Mode

The system automatically enters Track Mode following successful completion of BIT during Diagnostic or Maintenance Mode. Track Mode could be referred to simply as Surveillance. The two terms are interchangeable. However, in this document the term Track Mode will be used. Track Mode provides tracks of all aircraft with transponders within the surveillance volume. To exit track mode, the user must “stop” system software from the MIU.



Figure 4-14 System Mode/Integrity Monitor RCU Display

4.7.5 Active Guidance during Track Mode

The system enters active guidance during Track Mode when the TLS operator enters a valid transponder code from the RCU and clicks “Acquire ILS” or “Acquire GCA” [Figure 4-14]. During guidance, the system continues to provide surveillance track functions. The three guidance states are bulleted below:

- Acquisition
- Localizer-Only
- Tracking

4.7.6 Tracking

The TLS interrogates the service volume to establish surveillance tracks of aircraft responding with transponder ID codes. If approach guidance is requested through the acquisition of a specific transponder code and the desired aircraft is tracked to within Category I accuracy, the system establishes either Localizer-Only state or Tracking state guidance.

The Localizer-Only state indicates that the guidance track is not within the glide slope guidance volume and that only localizer guidance transmissions will be available.

Tracking state indicates that the tracked aircraft’s position estimate meets Category I guidance requirements for both localizer and glide slope, and both uplink guidance signals for localizer and glideslope are provided. Upon successful completion an approach or when the aircraft leaves the service volume, the software automatically stops active guidance mode.

4.7.7 Exiting Active Guidance

Guidance may also be exited via a RCU Reset command, which results in the system stopping the active guidance. If a track is dropped due to Integrity Monitoring alarms (e.g., the track no longer meets Category I accuracy), the system also stops active guidance. Depending on the alarm, guidance will resume automatically within seconds or may remain off. When the system exits active guidance, guidance radiation to the aircraft ceases and the pilot will see flags on the CDI/HIS display.

4.7.8 Test Mode

The system enters Test Mode when selected from the MI. During this mode, the Integrity Monitoring executive function is disabled. From this mode the TLS Technician may enable/disable BIT, schedule BIT at a temporary schedule for test purposes, command an immediate BIT, and start a test guidance track. Integrity monitoring information will be logged but the system will not discontinue guidance as a result of the monitor response.

4.7.8.1 Test Mode: Two States

Test Mode has two states: Test Track and Test Diagnostic. During Test Track active guidance is provided with these exceptions: 1) the Integrity Monitor is disabled; 2) the Morse Ident is removed and 3) the input of a transponder code can only come from the MI and never from the RCU. Guidance from Test mode is only provided for test purposes and should never be used in conditions other than visual meteorological conditions (VMC). The Test active guidance is aborted from an Abort command from the MI, and Test mode is exited upon receipt of an Exit Test Mode command from the MI.

4.8 The TLS Integrity Monitor status

The TLS Integrity Monitor (IM) [Figure 4-15] is designed to ensure the system provides guidance within the ICAO CAT I limits, and to provide system availability information prior to guidance, maintenance, and troubleshooting. The IM operates in 3 modes:

- Diagnostic
- Track
- Maintenance

The Diagnostic Mode BIT happens at start-up and at regularly scheduled intervals, after which the software transitions to Track Mode if no failures of the systems surveillance functions are detected. The IM in Track Mode provides two levels of availability: surveillance only and surveillance and guidance. The surveillance-only condition without guidance availability indicates that there is a failure with guidance-specific equipment such as the GTU and precision uplink guidance cannot be transmitted. Any failure in the surveillance functions during Track Mode will result in a transition to the Maintenance Mode. A failure to guidance equipment will require maintenance as well, and the system will indicate a failure, but the surveillance function remains operable until a maintenance technician shuts the system down for resolution.

The Cal/BIT pulse is received at the sensors is compared by the base station software against the monitor limits to ensure that the Cal/BIT generator and AOA sensors are operating within tolerance. The IM also monitors the guidance uplink, the health of the interrogation system, base station temperatures and other measurements required to ensure that the TLS is performing within specification.



Figure 4-15 Integrity Monitor Display on RCU

4.8.1 The Monitor Architecture

The IM design includes dissimilar, redundant CPUs and a failsafe compare of the track solutions implemented in the GTU(s). The CPUs are independent from the GTU and provide the GTU with data.

Likewise, the system follows this dissimilar IM design for the guidance data being transmitted to the RCU. The dual compare is done at the RCU to ensure the same level of quality data is delivered to the RCU for display during GCA. Inside the RCU, both sets of data from the two base CPUs are over-laid on the GCA display so the operator performing the GCA is able to detect any variance due to the RCU computing.

The IM design incorporates a monitor test signal transmitter called the Cal/BIT. The Cal/BIT transmitter consists of a fixed transmitter and associated antenna with a signal received by each AOA subsystem. The Cal/BIT transmitter provides a static source that can be used to detect AOA subsystem failures. Any aspects of the AOA subsystem that are non-linear and likely to be affected differently by a dynamic aircraft transponder than the static Cal/BIT source were identified during the design process. Monitor tests were devised to ensure that all possible degradations and failures are detected.

The monitors included with the localizer and glide slope uplink sample a small portion of the outgoing signal and demodulate it with a temperature-compensated envelope detector. This data is sent back to both CPUs for comparison to the commanded values. A hardware watchdog timer shuts off the transmitter output signal if the CPU(s) control inputs become inactive during guidance. The GTU carrier signal is monitored through a feedback loop from a signal pickoff.

4.8.2 Diagnostic BIT

Diagnostic BIT is run when the system is first powered on and periodically when the system mode is in Track or Maintenance. If one or more critical parameters are at or above the monitor alarm limit, the system mode will transition to Maintenance. If all monitor tests are within alarm limit, the system mode will transition to Track. If one or more tests are at or above the alert limit, the system mode will transition to Track, but will give an indication of a possible maintenance condition. The BIT cycle interval was determined from consideration of exposure time in the System Safety Analysis (SSA).

4.8.3 Integrity Monitoring During Track Mode

There are three Track Mode monitors. They are called Track BIT, GTU BIT and Periodic Diagnostic BIT. Each BIT cycle interval was determined by computing the maximum allowed exposure time during the System Safety Analysis (SSA).

- Track BIT is designed to monitor data that is important to the tracking or measurement functions of the system. This includes surveillance as well as guidance.
- GTU BIT is periodic test designed to assess the status of the guidance uplink transmitter and its output.
- Periodic Diagnostic BIT is a once-a-day full system Diagnostic BIT that can be scheduled by the operator to run during a known down period, normally midnight. Periodic Diagnostic BIT exercises all of the Track BIT, GTU BIT and includes other operationally interfering tests (Tilt, DDM compare, etc.) that cannot be done while the system is in Track Mode.

4.8.3.1 Track BIT

Track BIT will perform some of the tests performed in Diagnostic BIT, except for the tilt sensor self-test. BIT tests will run continuously by interleaving BIT interrogations with surveillance and guidance interrogations. Track BIT will only be performed in Track Mode. It will not be performed in the Track state of Test mode.

4.8.3.2 GTU BIT

GTU BIT will perform the same tests as in Diagnostic BIT, except for the DDM compare and T0 timeout tests. GTU BIT will not run for a GTU that has a fatal failure already detected in Diagnostic BIT. GTU BIT will run independent of Track BIT monitoring. It runs periodically, based on a configurable setting in the maintenance interface.

4.8.3.3 Periodic Diagnostic BIT

The term Diagnostic BIT refers to BIT performed in Diagnostic and Maintenance Modes and the Test state of Test mode. The tests are identical. Diagnostic BIT will be entered automatically from Track Mode once a day, at a time specified by a setting.

4.8.4 Integrity Monitoring During Guidance

Active guidance is a system state that occurs during Track Mode, which consists of three functional sub-states: Acquire, Localizer-Only, and Tracking.

During Acquire, no guidance to the aircraft is transmitted. During Localizer-Only, only localizer guidance is transmitted. During guidance state Tracking, both localizer and glide slope guidance is transmitted. The phrase active guidance is used below to include Localizer-Only and Tracking states.

IM tests are performed on every interrogation during active guidance. Some monitor tests and monitor limits during active guidance are based on whether the track resolution is "Low" or "High". For the ESA, the track resolution is high if the aircraft is within 1° of glide path, otherwise the track resolution is low. For the ASA, the track resolution is low from the time guidance is first provided until the aircraft is within 15° of centerline. After this, the track resolution is high and remains high, even if the aircraft leaves the volume within 15° of centerline. Note that the resolution will always be high if guidance is first provided within 15° of centerline.

The specified tolerances on the TLS signal in space have a limit on error magnitude and on error duration; a maximum angular error and a maximum allowed time that the error may exceed the prescribed maximum angular error. A means for characterizing the amount of error in terms of magnitude and duration are part of the TLS monitor architecture and are called Data Quality Monitor (DQM) and Track Monitor.

4.8.4.1 Data Quality Monitor

Most of the active guidance monitors measurement data tests apply a DQM to ensure that the measurements used in the track solution are accurate. Failure of a DQM test occurs when a critical parameter is equal to or exceeds that parameter's alarm limit. As a consequence of failing a DQM test, the measurement is tagged as invalid so it is not used in the computation of the track solution. The failure of other active guidance Track Monitors, usually the Update Rate tests, will result in an alarm that terminates guidance if there are too few measurements to compute an accurate solution.

Generally, the failure of a DQM that tests the Low antenna will only lead to an alarm if the track resolution is "Low". The failure of a DQM that tests the Medium or High antenna will only lead to an alarm if the track resolution is "High".

4.8.4.2 Track Monitor (TM)

The Track Monitor tests ensure that the guidance tracking algorithm has at a minimum a 5 Hz update rate, the track is reasonable, the track is stable, and the total period of radiation outside the performance limits does not exceed the Annex 10 requirements for a CAT 1 approach. Failure of any of these tests will cause the guidance approach to drop and uplink signals to cease. The RCU status information will provide an indication to the TLS Operator.

4.8.5 Log Files

IM results are stored in the following files:

- Track and active guidance alarms are stored in *base.log*.
- ESA BIT results are stored in *ESA.bit*.
- ASA BIT results are stored in *ASA.bit*.

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- ATA BIT results are stored in *ATA.bit*
- Uplink BIT results are stored in *uplink.bit*.
- Interrogation BIT results are stored in *interrog.bit*.
- Base CPU Disk Space and ASA / ESA CPU Time test results are stored in *cpu.bit*.

There are separate *.bit files for Track and Diagnostic Mode data stored in separate sub-folders.

4.8.6 Periodic Maintenance of the Integrity Monitor

The TLS has been designed to comply with ICAO integrity levels for a CAT I ILS. No periodic maintenance of the integrity monitor is necessary as a consequence of the rigorous system design analysis and safety assessment. Periodic flight inspection and analysis of BIT data will reveal corrective action to reduce failure of equipment that may be operating near its monitor tolerance limits. Periodic inspection of the BIT data may lead to some corrective action to circumvent an alarm condition. See document ANPC # 020-00074, the Maintenance Manual on Maintenance for BIT data processing and periodic Flight Inspection tolerances.

4.8.7 Alarm & Alert Conditions

The following Alarm and Alert scenarios are possible:

- Alarms that occur during active guidance are caused by degradation in system accuracy or performance. These alarms will drop the approach and cease guidance uplink. Certain failures may be specific to providing ILS guidance and, therefore, the system may continue to operate in Track Mode and provide GCA guidance.
- Alarms that occur during Track Mode are caused by degradation in system accuracy, performance, or hardware failure. These alarms indicate a failure that prevents the system from providing surveillance as well as guidance, and will transition the system into Diagnostic Mode. The Diagnostic Mode BIT determines whether a temporary failure or a hardware failure warrants system shutdown.
- Alarms that occur during Diagnostic or Maintenance Mode detect system accuracy or performance degradation due to a hardware or software system element failure. The system will transition to or remain in Maintenance Mode until corrected and the system will remain unavailable for both surveillance and guidance functions until the problem is corrected. BIT will continue to run during Maintenance Mode.
- Alerts will be issued to the TLS Operator and Maintenance Technician as a warning that critical signal parameters may be hovering near monitor alarm limits. To ensure high continuity of TLS service, the system will remain operational until the TLS Maintenance Technician takes the system down for corrective action. If no corrective action is performed, and the condition persists and worsens, an alarm may occur rendering the system inoperable.

SECTION 5: OPERATION

5.1 Electronics Rack

See the TLS electronics rack part number drawings 920-00704-xxx where xxx corresponds to the part number suffix provided with the system. Figure 5-1 below shows a typical single rack configuration with four GTU's.

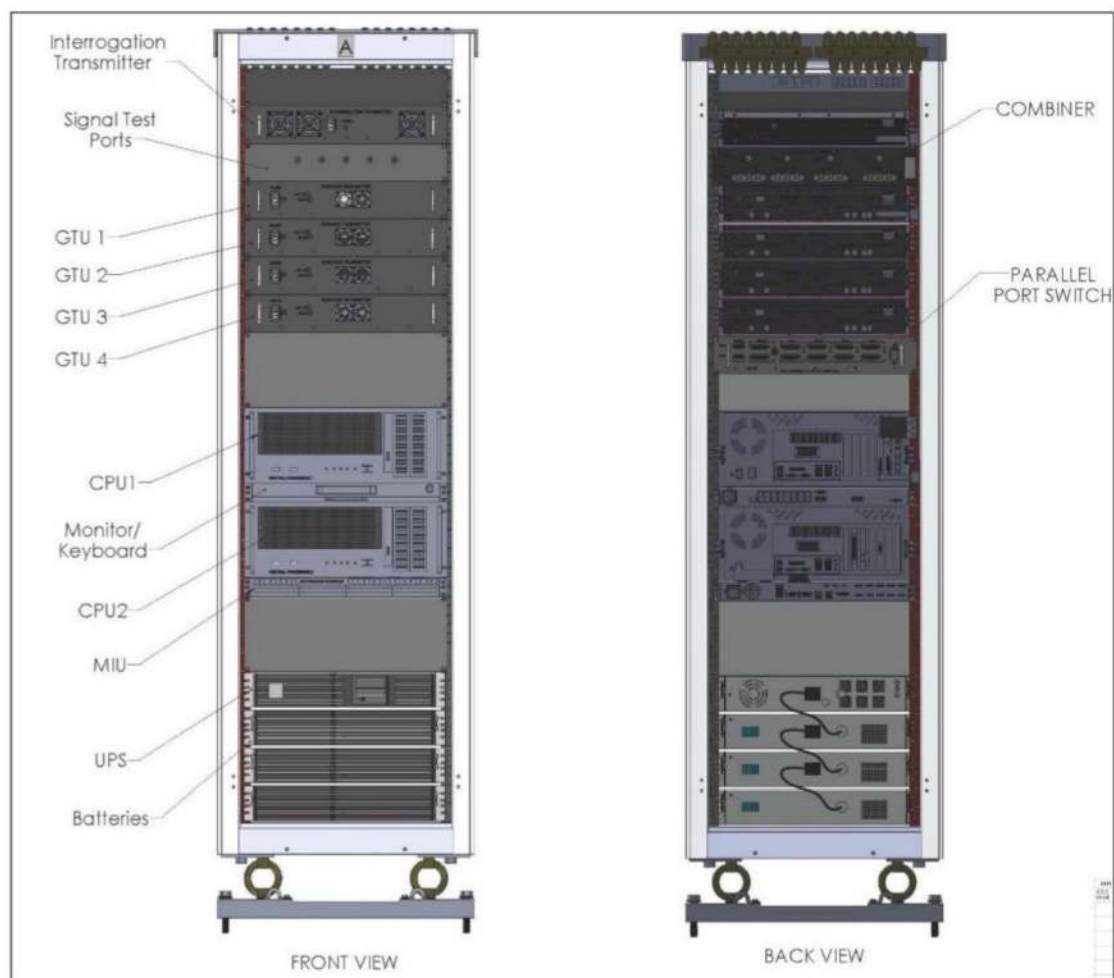


Figure 5-1 – example TLS Base Electronics Rack

5.2 Equipment Power Up/Down Procedure

Use the following steps to power up the TLS to the Track Mode of operation and power down the TLS to a cold (no power applied) state.

5.2.1 Power up Procedure

The base station keyboard, mouse and monitor are connected to the CPUs through a KVM switch, which is part of the keyboard/monitor tray.

Power up the TLS using the following sequence:

1. Place all circuit breakers in the ON position at the base shelter main service panel.

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2. Turn the Main and RSC UPS to ON to provide conditioned and protected power. The GXT3 UPS has a single power on switch. The GXT4 has an LED display and menu, follow the menu instructions to power on the GXT4
3. Turn the KVM monitor ON.
4. Turn the RSC ON. (if system is dual rack configuration)
5. Switch the monitor to the MIU. Turn the MIU ON and wait for the computer to run through the boot sequence. Login as "TlsTech" with the password tech1234.
6. Switch the monitor to CPU1. Turn CPU1 ON and wait for the computer to boot. When prompted, login as "root" with the password root.
7. Switch the monitor to CPU2. Turn CPU2 ON and wait for the computer to boot. Login as "TlsTech" with the password tech1234.
8. Turn the GTUs ON.
9. Turn the Interrogation Transmitter ON.
10. If both sensors are known to be on the network then go to step 13.
11. Select CPU1 and type ping ASA, then verify proper response. Press Ctrl-C to stop ping.
12. Type ping ESA then verify proper response. Press Ctrl-C to stop ping. The ping test can also be completed from a command prompt window on the MIU or Base2.
13. The electronics are now ready for TLS software to run.

5.2.2 Start TLS Base Station Software

To start the TLS software, perform the following steps.

1. At the Base Station, on the MIU, double-click the MI shortcut on the desktop to start the MI software.
2. In the MI main window, press the Start button. The MI system mode should display transitions from "No Comm with Base SW" to "Initializing," to "Diagnostic," and after completion of BIT, it should move either to "Track," or "Maintenance." Prolonged "Initializing" most often indicates that an LRU is not connected to the network; check that all LRUs are registered on the network (for example steps 11 and 12 above).

5.2.3 Start RCU Software

1. At the RCU, turn monitor and computer power switches ON. Wait for the RCU computer to boot. Login as "RCUOperator" with the password RCU1234. Verify that the keyboard, pointer, and monitor are working properly.
2. The RCU software should start automatically since it should be in the windows startup folder. There is an RCU icon on the desktop for restarting RCu software.

5.2.4 Power down Procedure

1. Switch the keyboard and mouse to the MIU. On the MI, press the Stop button to stop the system software. Shut down the MIU following normal Windows shutdown procedures. Turn the AC power OFF on the MIU.
2. Turn the AC power OFF on the Interrogation and GTU Transmitters.
3. Switch the keyboard and mouse to CPU2 and shut down following normal NT shutdown procedures. Turn the AC power OFF on CPU2.

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4. Switch the keyboard and mouse to CPU1 and type "sf -b" (shutdown script), and press enter. After the screen turns black, turn the AC power OFF on CPU1.
5. Turn off the RSC (if system is two rack configuration)
6. Turn off both UPSs.
7. Repeat steps 1 through 6 for the Reserve Rack (if two rack configuration)
8. Turn the main service panel breakers to the OFF position
9. Exit the RCU software, and follow normal windows shutdown procedure to shut down the computer. Turn off the monitor.
10. Exit the Remote Desktop Connection, and follow normal windows shutdown procedure to shut down the computer. Turn off the monitor.
11. The fans will continue to run on each UPS. If the system will be off for extended period of time or if it is desired that the fans remain off, unplug the UPSs from power receptacle on the wall and the UPS fans will stop within 30 seconds.

5.3 Operator Software Interfaces

The TLS has two interfaces: Maintenance Interface (MI) and the Remote Control Unit (RCU). Each interface is described below.

5.3.1 Maintenance Interface

The TLS maintenance technician uses the MI to access the TLS system status and perform diagnostics from the TLS base station. When the computer is started, the Windows login window will appear. The maintenance technician will enter a username and password.

The MI main window is displayed when the MI software is started [Figure 5-2], and contains three menus:

- System: contains items to start and stop the system software
- Tools: Discussion of tools begin in section 5.3.3 below
- Help: Discussion of this selection begins in section 5.3.4 below

The center of the MI main window has five buttons:

- Start: starts the system hardware
- Stop: stops the system software
- System Setup: opens the System Setup window
- System Status: opens the System Status window
- Archive: archives and cleans recorded data

The upper window text displays the runway number (read from the approach.dat file) and the current date and time.

CAUTION

If the UPS Status is in backup mode it will read ON BATTERY and display the number of minutes until system shutdown.

The lower window text displays the following: The System Mode (4) -Track, Diagnostic, Test, and Maintenance (Track Mode is shown in Figure 5-2). System Mode will display "NO COMM WITH BASE SW" if the TLS software is not running on CPU1 and CPU2 or if there is a network fault. If only one of the

Base CPU's software is not running while the other one is, the message "BASE X SW NOT COMMUNICATING" will be displayed (with X indicating 1 or 2 for CPU1 or CPU2).

The lower window text also shows the current UPS status; the current TRACK RAM file recording setting; and the current free disk space on the MIU.



Figure 5-2 Maintenance Interface Main Window

5.3.2 MI System Menu

This menu has three selections: Start System Software, which starts the TLS software on CPU1 and CPU2; Stop System Software, which stops the TLS software on CPU1 and CPU2; and Exit, which exits the MI application. Note that the Start and Stop TLS Software selections do not start or stop the sensor or RCU software.

5.3.3 MI Tools Menu

This menu's eight selections are described below.

5.3.3.1 Test Mode

CAUTION

This mode disables the Integrity Monitor and RCU. Moreover, in test mode there is no Morse Ident broadcast on the guidance. Issue a NOTAM prior to using Test Mode.

Figure 5-3 is displayed when Test Mode is selected from the system menu. Opening this window transitions the TLS to Test Mode, which prevents the RCU Operator from performing tracking operations.

Test Mode may only be started from Track Mode (if there are no active guidance tracks) and from Maintenance Mode. Test Mode is used to perform BIT during maintenance and troubleshooting efforts or to perform tracking operations without Integrity Monitoring enabled. The various parameters and command buttons are described below.

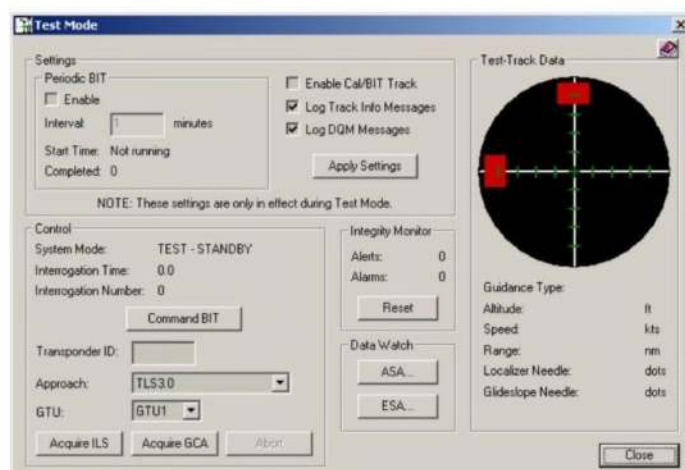


Figure 5-3 TLS Test Mode Window

Settings: Test Mode has its own settings, which allow temporary adjustment of system settings that are defined in settings.dat. The settings on the Test Mode Window are only in effect when the window is open.

- **Periodic BIT Enable:** When enabled, Periodic BIT sets up a Test Mode BIT cycle at the specified interval.
- **Enable Cal/BIT Track:** This selection enables the tracking of the Cal/BIT pulse. It is used to establish a track for troubleshooting if no aircraft are available. If this is set, the Transponder entry field below is automatically set to "0000" which tells the system to track the Cal/BIT pulse instead of a valid transponder code.
- **Log Track Info Messages:** enable logging of track info messages.
- **Log DQM Messages:** enable logging of DQM messages.
- **Apply Settings:** sends these settings to the Base CPUs for use during Test Mode. This must be done for the entered settings to take effect.

Control: This section of the window contains the buttons used to start a BIT, Command BIT, and to start a Test Mode track, Acquire. The Test Mode track uses the Transponder ID and Approach selected. The system mode is displayed read-only, as is the interrogation time, which is logged from the start of an approach.

Integrity Monitor: This section contains read-only displays of the number of alerts and alarms counted since Test Mode was started, or since the Reset button was pressed.

Test-Track Data: This section contains a read-only display of tracking parameters including altitude, speed, range, and localizer and glide slope needle corrections. It also contains a graphical CDI display of the needle positions.

Data Watch: This portion of the Test Mode window has test buttons to open up Data Watch Windows for each sensor, ESA and ASA. See Figure 5-4 and Figure 5-5 for examples of the ESA and ASA Data Watch windows. These windows are used to view raw phase, amplitude, TOA and tilt data from the sensors real-time during BIT or tracking. They are displayed when ESA or ASA buttons are selected from the Test Mode Window. They may be re-positioned on the screen to gain access to the Test Mode Window. Both Data Watch Windows may be displayed simultaneously. The data displayed is received real-time from the sensors during BIT or tracking interrogations. The display may be frozen using the Freeze Display check box at the top right. Each section of data is described below and differences between the localizer and glide slope are noted.

- **Cal/BIT:** This section displays Cal/BIT data received during BIT or during Test Mode-Acquire or Track state BIT. The timestamp indicates the sample time and the Cal/BIT present checkbox indicates whether or not the system has detected the Cal/BIT pulse.

Each antenna's raw phase measurements for Channels A and C are tabulated below. If an antenna/channel pair is not used as a measurement in the system, it will be displayed with an N/A and no data. The average power measurement for each antenna is also included in the table.

The TOA data for the near and far Cal/BIT pulse location is also included as are the Cal/BIT frequency and pulse width measurements.

For the ESA sensor, the tilt sensor measurements are also displayed.

- **Transponder:** This section displays transponder measurement data received once the TLS has successfully acquired and established a track on the desired transponder code. It includes transponder frequency, TOA, average power, and pulse width measurements.

The Close button closes the Data Watch Windows.

Figure 5-4 ESA Data Watch Window

Figure 5-5 ASA Data Watch Window

Close: This button closes the Test Mode Window and transitions the system from Test Mode to Diagnostic Mode to run BIT. All temporary parameters configured in Test Mode are discarded and the system configuration returned to those stored in settings.dat.

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5.3.3.2 GTU Test

The GTU Test Window [Figure 5-6] is displayed when "GTU Test" is chosen from the system pull-down on the MI Main Window. It is used to calibrate or verify the calibration of the GTU. The GTU Test tool is also used as a troubleshooting aid to isolate the GTU. The TLS software must be shutdown to use this tool, and when started, the tool queries the user to shut down the software, and if ok, starts the GTU Test tool. It displays GTU status information as well as control fields as described below:

- Localizer/Glide Slope: These duplicate sections contain information pertinent to the localizer or glide slope transmission of the GTU. They each contain three sections: Control, Monitor, and Status which contain columns for Base1 and Base2 information read back from each transmitter port through each base.
- DSA: this allows selecting monitor feedback attenuation to achieve the required monitor power
- Control: This section is used to command the localizer or glide slope transmitter and contains the Loc/GS On buttons, which send an enable to the transmitter; Enable and Reset control check boxes, which send these commands to the transmitter; and the DDM entry field, which allows for entry of commanded DDM. The DDM may be entered or chosen from the list. The Base1 and Base2 columns reflect the status of these items as read from the transmitter through the GTU port of each Base CPU.
- Monitor: This section contains data read from the localizer/glide slope transmitter from each Base. None of this information is enterable. It is updated with each read of the transmitter per the T0 cycle.
- Status: This section contains status information read from the localizer/glide slope transmitter from each Base. None of this information is enterable. It is updated with each read of the transmitter per the T0 cycle.
- Tone: This section contains information pertinent to the tone transmission section of the GTU. It contains two sections: Control, which contains columns for Control and Base1; and Monitor, which contains the monitored values for the commanded tone.
- Control: The Control column contains the On check box, which enables Tone transmission via Base CPU1. The Base1 column check box indicates the status of this command. The tone buttons below command each of the tones. The Command box contains the status of the tone selection read back from the transmitter.
- Monitor: The Monitor section contains the % Modulation for the commanded tone. It is also displayed in Counts.
- Health: This section contains read-only health values read from each Base CPU port on the transmitter. These values are updated per the T0 cycle during each read of the transmitter. They include all monitored GTU voltages and the GTU temperature as well as the T0 period OK and Frequency Programming Verify Status values.
- Display Format: This section contains three buttons for setting the units that data are displayed in: Hexadecimal, Decimal, or Engineering Units. It defaults to use Engineering Units.
- T0 Rate: This section contains four buttons for selecting the T0 or sampling rate: 20 Hz (50 msec), 10 Hz (100 msec), 5 Hz (200 msec), 1.1 Hz (909 msec), 0.9 Hz (1111 msec), or Off (no T0).
- System Status: This section contains a real-time status text display, which indicates the state of the tool, the network communication status to both bases, and calibration status information during calibration software execution.

The window contains a Close button that closes the window and exits the test tool. The TLS software may be re-started after the test tool exits.

- Calibration: This section contains entry fields for entering the PIR reading for localizer and glide slope SDM to use in the calibration. These are entered in % modulation as read from the PIR. The section contains two command buttons as described below:
 - ✓ Calibrate - This button executes the calibration software, which uses the input PIR SDM and 100 samples taken from the transmitter to determine average values to be used to calculate the monitor scale factors.
 - ✓ Monitor Scaling - This button opens the Monitor Scaling Window [Figure 5-7] which is used to calculate the GTU monitor scale factors and offsets from the calibration data produced above, which are written to the settings.dat file. When the window is first opened, the current calibration values are displayed. The Monitor Scaling window contains duplicate sections of data for localizer and glide slope transmitters as is described next.
- Input: This section has fields displaying the current calibration values including the carrier, the PIR SDM, and the monitor data read back for both the 90 Hz and 150 Hz at the volume edges. These values are used to calculate the scaling information below via the Calculate button.

If the PIR SDM value is not valid, an error message will occur since the software will be unable to properly calculate scaling values.

- Calculated Scaling: This section displays the calculated values for the ratio and offsets per the values entered above in the columns titled "Results". The Stored columns contain the data read back from the Base CPU2. The results column is enterable for testing purposes.

The window contains three command buttons. The "Send to Bases" button saves the calculated values to the settings.dat file and sends the file to both Base CPU1 and CPU2; the "Reload Cal Values" button refreshes the input values to those from the calibration if any changes have been made manually; and the Close button checks to see if the scaling information has been changed and saved to bases, and if not saved, queries the user to verify that they do not want to save the settings.dat. It also exits the window.

GTU Test

GTU1 | GTU2

Localizer (108.15 MHz)

Control: Loc On, Enable, Reset, DDM (0.0), Base1 (0.000), Base2 (0.000)

Monitor: Carrier (0.000), 90 Hz (0.000), 150 Hz (0.000), Reflected (0.000), SDM (0.000), DDM (0.000)

Status: CPU1 Provided DDM (checked), CPU2 Provided DDM (checked), DDMs Equivalent (0), Difference (0), Transmitter On, SWR OK, Monitor Freq. Lock, Transmitter Freq. Lock, A/D Overflow (4440), Reset, T0 (17218), DSA (0.0), 31.5, Int (18246), FPGA (7027), DSP (8047)

Glide Slope (334.55 MHz)

Control: GS On, Enable, Reset, DDM (0.0), Base1 (0.000), Base2 (0.000)

Monitor: Carrier (0.000), 90 Hz (0.000), 150 Hz (0.000), Reflected (0.000), SDM (0.000), DDM (0.000)

Status: CPU1 Provided DDM, CPU2 Provided DDM, DDMs Equivalent (0), Difference (0), Transmitter On, SWR OK, Monitor Freq. Lock, Transmitter Freq. Lock, A/D Overflow (5550), Reset, T0 (21330), DSA (0.0), 31.0, Int (22358), FPGA (C197), DSP (C0F7)

Tone

Control: Morse, A, B, C, On, Command (0)

Monitor: % Mod (0.00), Counts (0)

Health

Base1, Base2: T0 Period OK, Freq. Pgm. Verify

Monitor: Loc, GS, TX reg, RX reg, Dgtl reg, GS Temp (25.0), Loc Temp (25.0)

Calibration

LOC PIR SDM (40.0), GS PIR SDM (80.0), Calibrate, Monitor Scaling

Start Recording 0, Updates: 121, T0 Rate: 10.9, Use transmit calibration (checked)

Display Format: Hex, Decimal, Engineering (selected), T0 Rate (Hz): 0, 20, 10, 5, 1.1, 0.9, Off, System Status: Base 1 and 2 Ready, Close

Figure 5-6 GTU Test Window

GTU1 Monitor Scaling

Localizer		Glideslope																															
Input Carrier: 1850 PIR SDM: 40		Input Carrier: 1850 PIR SDM: 80																															
Monitor Readback -18 .18 90 Hz: 678 1788 150 Hz: 1788 678		Monitor Readback -20 .20 90 Hz: 1067 1779 150 Hz: 1779 1067																															
Calculated Scaling <table border="1"> <thead> <tr> <th></th> <th>Results</th> <th>Stored</th> </tr> </thead> <tbody> <tr> <td>90 Hz. Offset</td> <td>0.3333333</td> <td>0</td> </tr> <tr> <td>90 Hz. Ratio</td> <td>1</td> <td>0</td> </tr> <tr> <td>150 Hz. Offset</td> <td>0.3333333</td> <td>0</td> </tr> <tr> <td>150 Hz. Ratio</td> <td>1</td> <td>0</td> </tr> </tbody> </table>			Results	Stored	90 Hz. Offset	0.3333333	0	90 Hz. Ratio	1	0	150 Hz. Offset	0.3333333	0	150 Hz. Ratio	1	0	Calculated Scaling <table border="1"> <thead> <tr> <th></th> <th>Results</th> <th>Stored</th> </tr> </thead> <tbody> <tr> <td>90 Hz. Offset</td> <td>1.0000000</td> <td>0</td> </tr> <tr> <td>90 Hz. Ratio</td> <td>0.9993517</td> <td>0</td> </tr> <tr> <td>150 Hz. Offset</td> <td>1.0000000</td> <td>0</td> </tr> <tr> <td>150 Hz. Ratio</td> <td>0.9993517</td> <td>0</td> </tr> </tbody> </table>			Results	Stored	90 Hz. Offset	1.0000000	0	90 Hz. Ratio	0.9993517	0	150 Hz. Offset	1.0000000	0	150 Hz. Ratio	0.9993517	0
	Results	Stored																															
90 Hz. Offset	0.3333333	0																															
90 Hz. Ratio	1	0																															
150 Hz. Offset	0.3333333	0																															
150 Hz. Ratio	1	0																															
	Results	Stored																															
90 Hz. Offset	1.0000000	0																															
90 Hz. Ratio	0.9993517	0																															
150 Hz. Offset	1.0000000	0																															
150 Hz. Ratio	0.9993517	0																															
Reload Cal Values Send to Bases Close																																	

Figure 5-7 Monitor Scaling Window

5.3.3.3 GTU Frequency Programming

This window [Figure 5-8] is displayed when the GTU Frequency Selection is selected from the Tools menu on the MI Main Window. It provides a mechanism to program the EPROM on the GTUs that stores the ILS frequencies to transmit for both localizer and glide slope. For the 920-00101 GTU, the frequency programming jumper must be installed in the rear left upper corner with the lid removed to provide write access for the software. Each section of the window is described below:

- **Program Frequency:** The localizer frequency to program is input in the localizer Frequency pull-down list and the Program button is then used to send the frequencies to the GTU. The localizer and glide slope fields show the frequencies commanded to the GTU.
- **Read Frequency:** This section contains the frequency data read from the GTU via each Base CPU port. The data contains fields for the frequency read from each base (localizer and glide slope) and the contents of the registers. This data is updated when the frequency tool is started and when the Program button is clicked. It may also be refreshed using the Read button in this section.
- **Close** - This button closes the window.

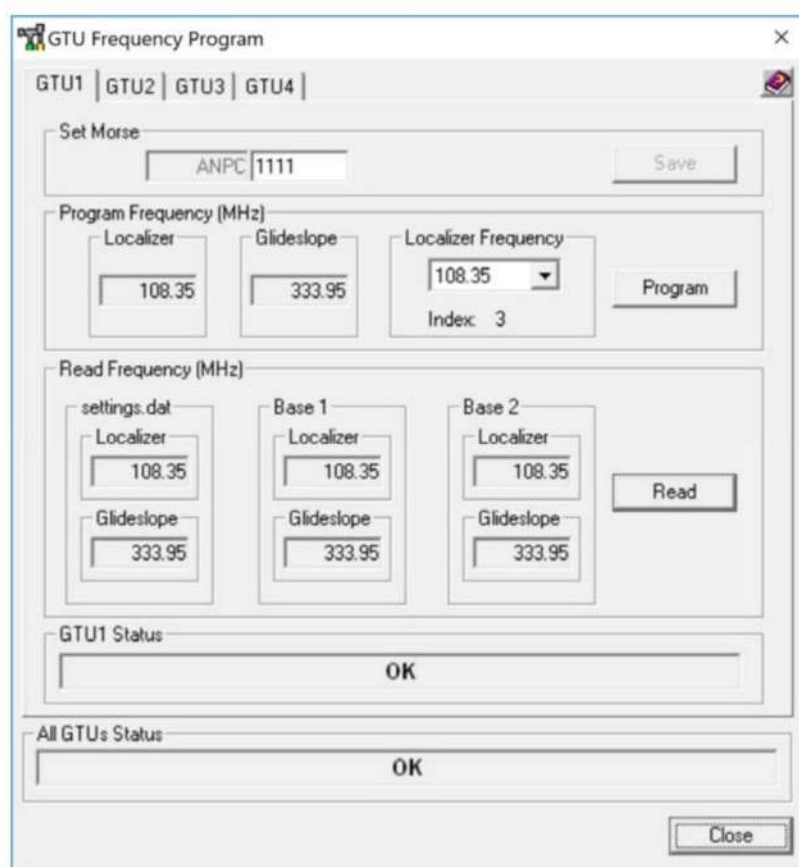


Figure 5-8 GTU Frequency Programming Window

5.3.3.4 Start FTM

This tool starts the Flight Test Monitor (FTM) application used to verify TLS performance against truth data recorded with a theodolite. This tool connects to the theodolite via a serial link to a modem.

5.3.3.5 System Data

The System Data Window [Figure 5-9] is displayed when System Data is selected from the MI Main Window Tools menu. It displays real-time guidance track information and may be opened during any mode. It is also movable on the screen so that it may be conveniently located. This window is useful for the Maintenance Technician if guidance tracks are being initiated from the RCU to view system status. Information includes:

- System Mode: The current TLS mode.
- Interrogation Time: The current interrogation timestamp.
- Integrity Monitor: The current count of alarms and alerts since the Track Data Window was opened or the Reset button was pressed.
- Track Data: Includes the current aircraft altitude, speed, range, and localizer and glide slope needle corrections. It also contains a graphical CDI display of the needle positions.

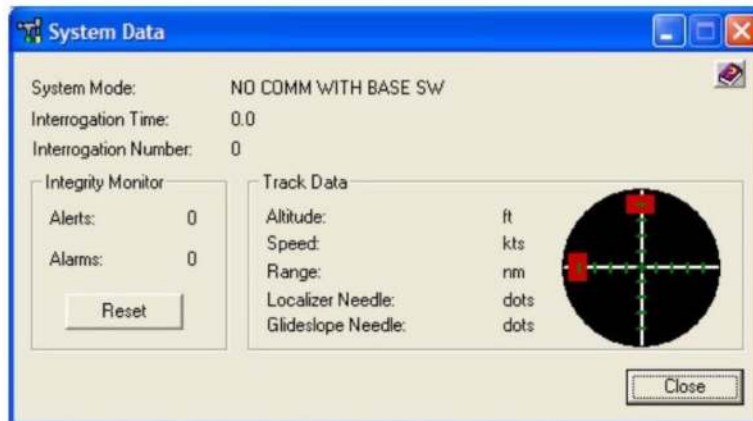


Figure 5-9 System Data Window

5.3.3.6 Archive

This starts the Archive tool [Figure 5-10]. This tool retrieves all data from the Base CPUs and zips into a file located on the MIU, in the archives directory. It then deletes all the data on both Base CPUs.

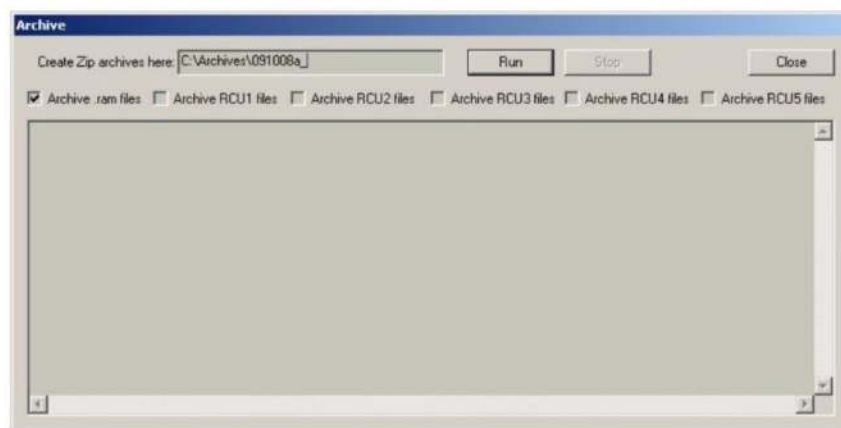


Figure 5-10 Archive Window

5.3.4 Help

This menu has two selections:

- Help: under development;
- TLS Version Information, which opens the TLS Version window.

5.3.4.1 MI Help

This option is under development.

5.3.4.2 System Information

The System Information window is displayed when System Information is selected from the Help menu on the MI Main Window. It displays the current software version. Also included is the software build date and timestamp. This information can be used to verify Configuration Items and installation procedures.

5.3.5 System Setup Window

The System Setup Window [Figure 5-11] is displayed when the System Setup button is selected on the MI Main Window and is used to set TLS system limits and other site configuration information. This window

should only be used by the TLS maintenance technician. The window contains four buttons to navigate to each part of System Setup. These buttons are: Monitor Limits, Survey, Settings, and ILS Settings. The text next to each of these buttons contains the time and date stamp of the latest version of the files for reference. Each of these will be described in following paragraphs.

The Close button closes the window. If file changes have occurred and they have not been saved to the Base CPUs, the user will receive a warning and a chance to stop and send them.

Each of the windows available from the System Setup has the same three buttons that behave in the same way so the descriptions below apply to all of them.

OK - This button closes the window and also saves any changes to the corresponding configuration file (either monitor_limits.dat, settings.dat, or survey.dat). When this button is pressed the software prepares to send configuration files to both CPU1 and CPU2. If the TLS software is running on either or both of the Bases, the MI will notify the user and ask whether or not they want to shut the software down.

The TLS software only reads the files upon start-up so the software must be restarted for file changes to take effect. Once the software is shut down any calculation required are performed (i.e., calculating new limits based on new nominal values), the file checksum is computed, saved to the file and the file is sent to both Base CPUs; the user is notified of errors if any are encountered.

Apply - This button saves the data to the corresponding configuration file (either monitor_limits.dat, settings.dat, or survey.dat), sends the file to the bases (see OK above) but leaves the window up. Pressing the Apply button followed by the OK button has the same effect as pressing the OK button.

*Changes saved using the Apply button may not be erased using Cancel.

Cancel - This button closes the window without saving any changes that have not already been saved via Apply. If changes have been made and not saved via Apply or OK, the user receives a warning and can either return and save the changes or continue to Cancel and abandon the changes.

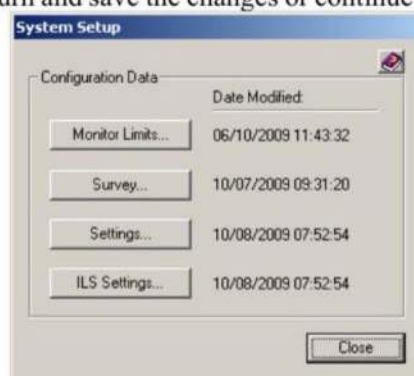


Figure 5-11 System Setup Menu

5.3.5.1 Monitor Limits Window

This window [Figure 5-12] is displayed when the Monitor Limits button is selected from the System Setup Window. It contains five buttons as shown below:

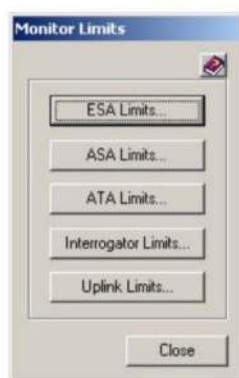


Figure 5-12 Monitor Limits Window

5.3.5.1.1 ESA Limits:

Pressing this button displays the ESA Monitor Limits window [Figure 5-13]. Entry fields are available for each value that determines a TLS monitor limit. The data is saved in the monitor_limits.dat file.

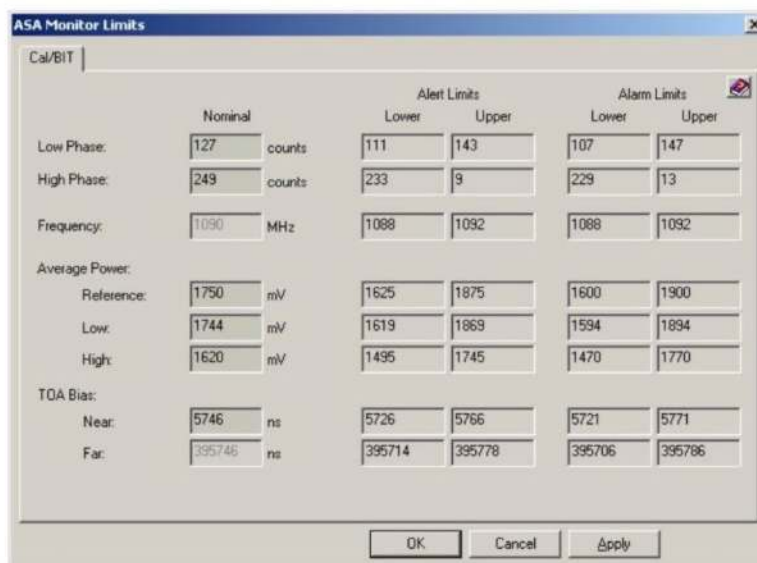
The Tilt Sensor tab contains entry fields for nominal tilt parallel to and perpendicular to the runway. Each of the tabs contains the same three buttons - Apply, Cancel, and OK.

	Nominal		Alert Limits		Alarm Limits	
			Lower	Upper	Lower	Upper
Low Phase:	4 counts		244	20	240	24
Medium Phase:	88 counts		72	104	68	108
High Phase:	254 counts		238	14	234	18
Frequency:	1090 MHz		1088	1092	1088	1092
Average Power:						
Reference:	1650 mV		1525	1775	1500	1800
Low:	1474 mV		1349	1599	1324	1624
Medium:	1509 mV		1384	1634	1359	1659
High:	1420 mV		1295	1545	1270	1570
TOA Bias:						
Near:	6421 ns		6401	6441	6396	6446
Far:	396421 ns		396389	396453	396381	396461

Figure 5-13 ESA Monitor Limits Window

5.3.5.1.2 ASA Limits:

This button opens the ASA Monitor Limits Window [Figure 5-14]. Entry fields are available for each value that determines a TLS monitor limit. The data is saved in the monitor_limits.dat file.



The ASA Monitor Limits window displays various parameters and their limits. The parameters are organized into rows, each with a label, a nominal value, a unit, and two columns for alert limits (Lower and Upper) and two columns for alarm limits (Lower and Upper). The parameters include Low Phase, High Phase, Frequency, Average Power (Reference, Low, High), and TOA Bias (Near, Far).

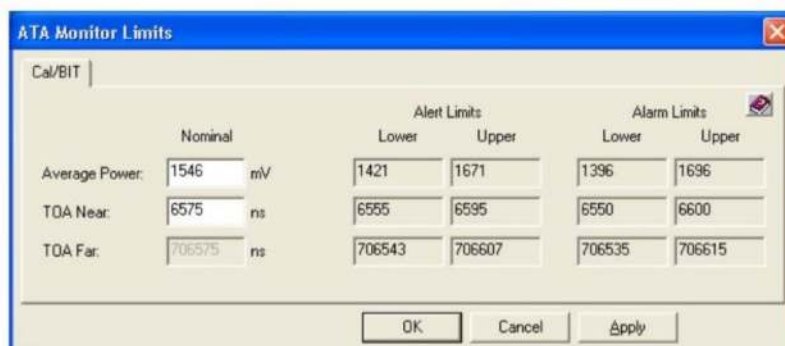
	Nominal		Alert Limits		Alarm Limits	
			Lower	Upper	Lower	Upper
Low Phase:	127	counts	111	143	107	147
High Phase:	249	counts	233	9	229	13
Frequency:	1090	MHz	1088	1092	1088	1092
Average Power:						
Reference:	1750	mV	1625	1875	1600	1900
Low:	1744	mV	1619	1869	1594	1894
High:	1620	mV	1495	1745	1470	1770
TOA Bias:						
Near:	5746	ns	5726	5766	5721	5771
Far:	395746	ns	395714	395778	395706	395786

Buttons: OK, Cancel, Apply

Figure 5-14 ASA Monitor Limits Window

5.3.5.1.3 ATA Limits:

This button opens the ATA Monitor Limit Window [Figure 5-15]. The window displays the ATA nominal value, alert and alarm limits for the Cal/BIT average power Near TOA and Far TOA.



The ATA Monitor Limits window displays parameters and their limits. The parameters are organized into rows, each with a label, a nominal value, a unit, and two columns for alert limits (Lower and Upper) and two columns for alarm limits (Lower and Upper). The parameters include Average Power, TOA Near, and TOA Far.

	Nominal		Alert Limits		Alarm Limits	
			Lower	Upper	Lower	Upper
Average Power:	1546	mV	1421	1671	1396	1696
TOA Near:	6575	ns	6555	6595	6550	6600
TOA Far:	706575	ns	706543	706607	706535	706615

Buttons: OK, Cancel, Apply

Figure 5-15 ATA Monitor Limits

5.3.5.1.4 Interrogator Limits:

This button opens the Interrogator Limits Window [Figure 5-16], which contains entry fields for nominal P1/P3 amplitude values and P2 amplitude values. Note that if P2 is not enabled, this nominal does not need setting since there is no monitoring of this parameter. This window also contains the Apply, Cancel, and OK buttons. The data is saved in the monitor_limits.dat file.

	Nominal	Alert Limits		Alarm Limits	
		Lower	Upper	Lower	Upper
P1/P3 Nominal Amplitude:	550 mV	436.9	N/A	389.4	N/A
P2 Nominal Amplitude:	550 mV	436.9	N/A	389.4	N/A

Buttons: OK, Cancel, Apply

Figure 5-16 Interrogator Monitor Limits Window

5.3.5.1.5 Uplink Limits:

This button opens the Uplink Limits Window [Figure 5-17], which contains entry fields for nominal localizer and glide slope carrier and reflected power values. Note that these values are expected in mV. The Uplink Limits window also contains the Apply, Cancel, and OK buttons. The data is saved in the monitor_limits.dat file.

	Nominal	Alert Limits		Alarm Limits	
		Lower	Upper	Lower	Upper
Localizer:					
Carrier Sample Power:	1850 mV	1469.507234	N/A	1309.699702	N/A
Reflected Power:	550 mV	389.3701819	776.8956490	N/A	N/A
Glide Slope:					
Carrier Sample Power:	1800 mV	1429.790822	N/A	1274.302411	N/A
Reflected Power:	500 mV	353.9728919	706.2687720	N/A	N/A

Buttons: OK, Cancel, Apply

Figure 5-17 Uplink Monitor Limits Window

5.3.5.2 Survey Window

This window [Figure 5-18] is displayed when the Survey button is selected from the System Setup Window. It contains entry fields for TLS antenna locations. The expected units are listed next to the entry box. The values to enter are determined during the survey procedure. Note that all angles are entered in decimal degrees-minutes-seconds format, range distances are in feet and antenna separations are in inches. The data is saved in the survey.dat file.

The Survey Window dialog box contains the following sections and fields:

- ASA:**
 - Reference Antenna: Range: 661.9 feet; Azimuth Angle: 169° 6' 52"
 - Separation from Reference Antenna: Low Antenna: 30 inches; High Antenna: 120 inches
- Cal/BIT Antenna:**
 - Range: 382.2 feet; Azimuth Angle: 130° 51' 27"
- Interrogator Antenna:**
 - Range: 752.0 feet; Azimuth Angle: 149° 48' 53"
- Main Runway:**
 - MSL altitude at threshold: 0.0 feet
 - Runway heading: 45.0°
 - Theodolite Eyepiece and Prism staff vertical offset: 5.5 feet
- ESA:**
 - Tower Base Plate (Reference Antenna): Range: 792.3 feet; Azimuth Angle: 145° 7' 17"; Elevation Angle: + 0° 0' 0"
 - Tower Base Plate Separation from Reference Phase Center: 62.170 inches
 - Phase Center Separation from Reference Antenna: Low Antenna: 89.00 inches; Medium Antenna: 178.00 inches; High Antenna: 267.00 inches
- ATA:**
 - Range: 977.8 feet; Azimuth Angle: 162° 48' 24"
- FTM Theodolite location:**
 - Range: 0.00 feet; Azimuth Angle: 0° 0' 0"; Elevation Angle: + 0° 0' 0"

Buttons: OK, Cancel, Apply

Figure 5-18 Survey Window

5.3.5.3 System Settings Window

This window [Figure 5-19] is displayed when the Settings button is selected from the System Setup Window. It contains entry fields for setting various system parameters, which are explained below. Note that units for these settings are listed next to each field. The data is saved in the settings.dat file.

1. Diagnostic Mode BIT Schedule:

- There are three BIT intervals that may be set in Diagnostic Mode BIT Schedule section. The first, Diagnostic BIT Schedule, is used to set the time period and number of retries that the system will perform upon BIT failure in Diagnostic Mode. The daily Diagnostic schedule is also set here.
- The second, Maintenance Mode BIT Schedule, is used to set the cycle that BIT is run from Maintenance Mode.
- The third, GTU Track BIT schedule is used to set the time between GTU BITs conducted during Track Mode. Setting this to 60 minutes reduces the occurrence of brief uplink signal transmissions that can be detected by any tuned ILS receivers in the area.

2. F1 Cal/BIT Position:

This is used to set the amount of time to delay the Cal/BIT pulse following the start of the T-Zero. This is used during system configuration to move the Cal/BIT pulse to the time position with minimum TOA jitter.

3. UPS Shutdown:

When the UPS is running from battery power and the remaining battery duration has been depleted to the number of minutes indicated in this field, the TLS software will begin the graceful shutdown of the CPUs and sensors.

4. Environment:

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- The Barometric Pressure field is set to standard altimeter sea level pressure 29.92. This causes the altimeter display next to each surveillance track on the RCU to be QNH (query: nautical height), the height above sea level when corrections are applied for local atmospheric pressure that is above or below the standard altimeter. Note that this field could alternately be set to the current field elevation pressure for QFE (query: filed elevation) if surveillance track altitudes relative to the airport/field elevation are desired.
 - The Transition Level field is the altitude at which the system transitions to universal pressure.
 - The Noise Level field is used to set a minimum amplitude level above which the sensor will detect pulse data.
 - The Compress Sensor Data field is used to control compression of data sent from the sensors. This is a read-only field.
5. Service Volume:
- These values define the TLS service volume where localizer and glide slope guidance signals will be received by the aircraft.
 - For TLS localizer guidance to be active, the aircraft must be within the horizontal area defined by the localizer Range and Azimuth Volume Edges.
 - For TLS glide slope guidance to be active, the aircraft must be within the azimuth volume defined for the glide slope and within the glide slope Range and Elevation Volume Edges
 - Once localizer guidance is active, if the aircraft leaves the localizer volume, guidance will be disabled. Once glide slope guidance is active, if the aircraft leaves either the localizer or glide slope volumes, guidance will be disabled.
6. Interrogation:
- This is used to set the interrogation rates for both surveillance and guidance in Hz. The guidance rate is nominally set to 10 Hz and the surveillance rate to 2 Hz. Checking the Enable P2 box enables side lobe suppression.
7. TOA Delay Settings:
- These settings are used to calibrate signal propagation times through antenna cables and fibers. Correct settings are required to achieve accuracy of the surveillance tracks. They are set during system installation.
8. ESA Tilt Sensor/ Diversity Antenna:
- The tilt sensor and diversity antenna enable/disable status. These are not configurable on the display and are shown so that the Maintenance technician can determine the system configuration if needed.
9. Data Recording:
- These parameters define various aspects of output file recording. The first character of the raw files can be set. Raw file recording can also be enabled. RAM file recording in both BIT and Track Modes can be enabled also as well as RAM recording during guidance only. Note that these settings are in effect until disabled and that output files can get quite large, especially RAM files recorded in Track Mode. GTU Files can also be recorded and various messages logged to the base.log can be enabled or suppressed.

System Settings

Diagnostic Mode BIT Schedule Retry Interval: 10 seconds Retries: 0 Daily BIT Time: 12:00:00 AM	Service Volume Az Volume Edges: +/- 42 degrees EI Volume Lower Edge: 0.9 degrees EI Volume Upper Edge: 7 degrees Localizer Range: 24 NM Glideslope Range: 12 NM EI Az Volume Edges: +/- 10 degrees	TOA Delay Settings ESA: 0 ns ASA: 0 ns ATA: 0 ns
Maintenance Mode BIT Schedule Interval: 5 minutes	Acquire Settings Acquire Timeout: 2 minutes Reacquire Limit: 0	Data Recording Record Track RAM Files: ☒ Off ☐ Guidance ☐ All ☒ Record BIT RAM Files ☒ Record Raw Files ☒ Record GTU Files ☒ Record Track BIT Files ☒ Log Track Info Messages ☒ Log DQM Messages First Character of Raw File Name: x Send FTM Data for this RCU: RCU1
GTU Track BIT Schedule Interval: 60 minutes	Interrogation Guidance: 10 Hz Surveillance: 2 Hz ☒ Enable P2	Rack Config Dual ☐ Single ☒ Rack A ☐ Rack B
F1 Cal/BIT Position Near: 5 μ s	UPS Shutdown 10 minutes	
Environment Barometric Pressure: 29.92 in Hg Transition Level: 18000 feet Max Noise Level: 40 counts ☐ Compress Sensor Data	GTU Type ☐ 920-00101 ☒ 920-00402 ☒ Multi-GTU (920-00706) Installed ☒ GTU 1 Installed ☐ GTU 3 Installed ☒ GTU 2 Installed ☐ GTU 4 Installed	

OK Cancel Apply

Figure 5-19 System Settings Window

5.3.5.4 ILS Settings Window

The ILS window [Figure 5-20] is displayed when the ILS Settings button is selected from the System Setup Window. It contains entry fields for enabling or disabling the GTU and for assigning RCU's to GTU's. The data is saved in the settings.dat file.

GTU	Approach Designator	RCU1	RCU2	RCU3	RCU4	RCU5	Select All	Clear All
GTU1	TLS VECCU 1 A 00	☒	☒	☒	☒	☒	Select All	Clear All
GTU2	TLS VECCU 2 A 00	☒	☒	☒	☒	☒	Select All	Clear All
GTU3	TLS VECCU 3 A 00	☒	☒	☒	☒	☒	Select All	Clear All
GTU4	TLS VECCU 4 A 00	☒	☒	☒	☒	☒	Select All	Clear All

Figure 5-20 ILS Setting Windows – will display controls for up to four GTUs

5.3.6 System Status Window

The System Status window is displayed when selected from the MI Window [Figure 5-2]. It displays the status of LRUs, retrieves base.log files from either base CPU, and can be used to verify network connections. The red arrow in the lower portion of Figure 5-21 points to the System Mode.

CAUTION

When the UPS button is yellow it means that the system is running on the UPS battery or there is no communications with the UPS.

On the MI computer the window can be opened in any TLS mode and may be moved on the screen and left open to access other items such as the Test Mode window. The graphical display and commands are described below:



Figure 5-21 System Status Window

1. **Graphical Display** - The graphical display contains status indicator boxes for all monitored physical LRUs including the ASA, ESA, ATA, Network, CPU1/2, RCUs, Interrogator, and the GTUs. It also displays the status for Track Monitor alarms and alerts generated during Track Mode. The LRU display uses different colors to represent different LRU states:

- **BLUE:** TLS software not running and no status is available.
- **GREEN:** The LRU does not have alarm or alert condition.
- **YELLOW:** The LRU has an alert condition.
- **RED:** The LRU has alarm condition.

The LRU images on the display can be clicked to access the base.log messages that pertain to that specific LRU. The content of this "base.log" file varies depending on what viewing mode is used as described below.

2. **View:** There are two viewing modes on System Status: Current and History. Current displays the current state of the system and is the default when the window is opened. The Current state is updated at the end of each BIT and is initialized to green at the start of a track or Test mode track so that alarms or alerts can be viewed real-time as they occur.

The History mode displays the state of each LRU based on a review of all alarms and alerts that exist in the base.log on CPU2. This can be useful if an intermittent alarm or alert is occurring but is not shown in the current state.

3. **Base1 Base Log:** This command button opens a window containing the entire base.log retrieved from CPU1.
4. **Base2 Base Log:** This command button opens a window containing the entire base.log retrieved from CPU2.

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5. **Base1 Startup Log:** This command button opens a window containing the sm.log from either CPU1, which contains start-up information. This button is only available when the TLS software is not running and is useful if the TLS software fails to start, since initialization errors are recorded here by the software.
6. **Base2 Startup Log:** This command button opens a window containing the sm.log from either CPU2, which contains start-up information. This button is only available when the TLS software is not running and is useful if the TLS software fails to start, since initialization errors are recorded here by the software.
7. **Network Test:** This command button opens the Network Test tool, which can be used to troubleshoot network connections. It contains three buttons: *Run*, *Stop*, and *Close*. *Run* performs an ICMP "ping" of each TLS host. The results of the ping tests are displayed in the window. Pressing *Stop* terminates the ping test. *Close* exits the Network Test tool and returns the user to the System Status window.

5.4 Operator Interface - Remote Control Unit

- The Remote Control Unit (RCU) consists of a computer, display [Figures 3-23, 3-24, 3-25, and 3-26], keyboard and mouse that are placed in a location convenient to the RCU Operator. The RCU Operator interface to the TLS is implemented in a single software program named RCUX.exe.
- The RCU allows the operator to replay previous track data. The replay function will only be available when the system software is not operational. The display is also tagged in the upper right hand portion with the label "Previously Recorded Data".

5.4.1 Interface Display

The system status and readiness is presented with visual (green, yellow, red) and audible (alarm sounds) cues to make it easy to discern operational conditions.



Figure 5-22 RCU Graphical Display in Yellow (alert) Mode

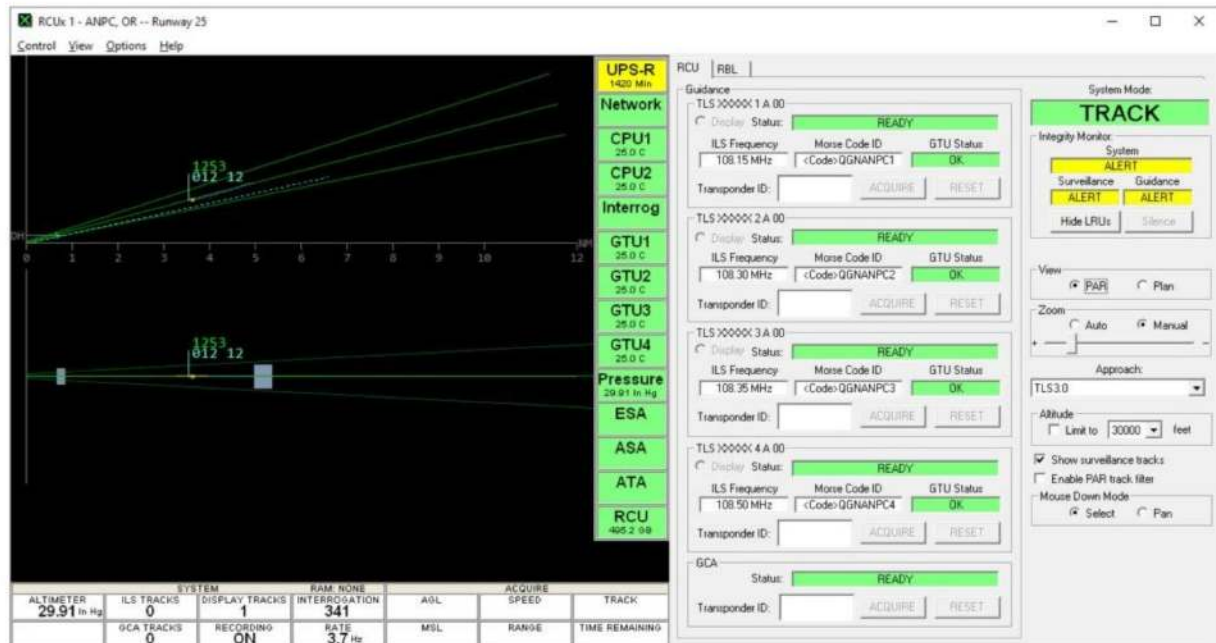
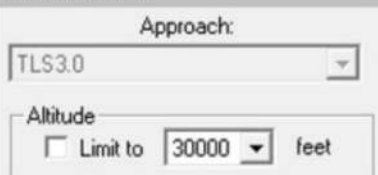
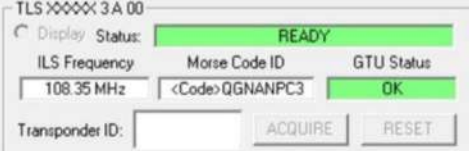
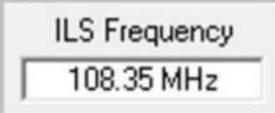


Figure 5-23 RCU Precision Approach Radar (PAR) Display

5.4.2 Interface Controls

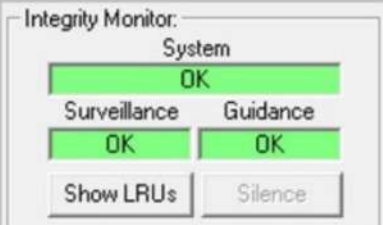
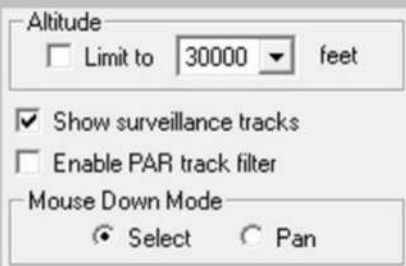
Table 5-1 contains a functional description of the status displays and controls that are provided on the RCU Operator Interface Screen.

Table 5-1 RCU Operator Interface Description

RCU/RBL Windows 	The RBL window can be used to draw Range Bearing Lines. A line is drawn from the start point to the end point. The start and end points can be a range and bearing or an aircraft.
View 	Select either plan view or PAR graphical display
Zoom 	Select manual and then set the range or set to automatic and the display will automatically adjust after the aircraft travels the distance listed under the Options menu.
Approach Window 	Select the desired approach profile. The TLS allows the RCU operator to select from among pre determined approach paths that have been designed using ICAO PANS-OPS and that have been flight inspected.
Guidance Window 	Each RCU has up to five guidance windows: four ILS guidance windows and one GCA window. This is where the RCU Operator interacts with individual aircraft approaches.
Transponder ID 	The Transponder ID window is used by the RCU Operator to enter the transponder code for the aircraft to be given guidance by the TLS.
ILS Frequency 	The ILS Frequency window indicates the assigned ILS localizer frequency for approaches conducted from this RCU. The window will be highlighted magenta if that approach is selected for display.
Status: 	White Screen - System software had not been started

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Status: READY	Green Screen - GTU is ready to use
Status: SEARCHING FOR CODE 1256	Yellow Screen - GTU in use. The TLS is searching for an aircraft and is in this state prior to providing guidance
Status: TRACKING LOC ONLY	Blue Screen - GTU in use. The aircraft has been identified and is being tracked by the TLS. Guidance is being provided for Localizer only.
Status: TRACKING	Blue Screen - GTU in use. The aircraft has been identified and is being tracked. Guidance is being provided for both Localizer and Glide slope.
Status: ILS GUIDANCE ACTIVE RCU2	Blue Screen - GTU in use. The aircraft has been identified and is being tracked from another RCU.
Status: GCA GUIDANCE ACTIVE	White Screen - GTU is unavailable. GCA guidance is being provided from this RCU.
Status: GUIDANCE SUBSYSTEM ALARM	Red Screen - GTU is unavailable. Guidance subsystem is in alarm and the Maintenance Technician should be notified.
Status: FAILED GTU BIT	Red Screen - GTU is unavailable. This indicates that ILS operations are not available from this GTU. The Maintenance Technician should be notified.
GTU Status GTU Status GTU Status OK ALARM	It will indicate a failed ILS transmitter with "ALARM" and be colored RED. This indicates that ILS operations are not available from this GTU. The Maintenance Technician should be notified.
ACQUIRE ACQUIRE RESET	ILS: Enabled only if TLS is ready for ILS guidance operations (and is not performing GCA guidance from this RCU) and has an operational ILS transmitter and frequency. Starts ILS guidance <u>within 3 to 5 seconds (make take longer)</u> for transponder code entered above. GCA: Enabled only if TLS is ready for GCA guidance operations (and is not performing ILS guidance from this RCU). Starts GCA guidance for transponder code entered above.
RESET	Enabled only if TLS is actively searching or providing guidance and stops guidance operations from this RCU. Use of this button will stop guidance and should be used with caution.
System Mode Status Window	The System Mode status window is located on the upper right side of the screen and indicates the current mode or interim states. The following are all possible indications that can be displayed and their meaning:
System Mode: TRACK Integrity Monitor: System OK Surveillance OK Guidance OK Show LRUs Silence	
NO COMM	Red - System NOT Available. The connection with the Base Station is broken or the system is not running. Maintenance Technician must be notified.
INITIALIZING	Blue - System in Use. The TLS is running initialization and startup routine.
DIAGNOSTIC	Blue - System in Use. The TLS is running BIT. A yellow screen indicates that an alert has occurred during BIT.
TRACK	Green - System Available. The TLS is ready to provide guidance and is providing surveillance.
MAINTENANCE	Red - System NOT Available. The TLS is not available for operations. The Maintenance Technician must be notified.
TEST	Blue - System in Use. The system is in Test Mode and cannot be controlled from the RCU.

Integrity Monitor Status Window	The Integrity Monitor status window, located below the System Mode status window, reports the status of the Integrity Monitor. Three possible indications are:	
	OK	Green - The TLS is healthy and ready to perform surveillance and guidance operations.
	ALERT	A Built-In-Test condition indicative of a non-critical TLS problem has occurred. Alerts do <u>not</u> affect TLS performance and do not require RCU Operator intervention. The alert can affect only guidance which indicates that a guidance-specific fault may occur and that guidance availability may be affected.
	ALARM	The TLS system (either surveillance or guidance) has degraded and is not available for standard operations. A guidance-only ALARM condition does not prevent surveillance operations but a surveillance ALARM prevents all operations. An audible alarm is also generated. The SILENCE button is enabled to quiet the audible alarm. The Maintenance Technician must be called.
Additional Controls 	Hide LRUs	This allows the RCU Operator to show or hide the LRU status bar on the graphical display.
	Altitude	Allows RCU Operator to limit the surveillance tracks displayed to those under the specified altitude.
	Show Surveillance Tracks	Allows the RCU Operator to disable display of surveillance tracks to clear the display during an approach if desired.
	Enable PAR Track Filter	Allows the RCU Operator to enable or disable the filter of the PAR surveillance tracks shown per the settings in the Settings Menu. This can help clear the PAR display during an approach.
	Mouse Down Mode	This allows the RCU Operator to change the mouse behavior to Pan to allow selective panning in the graphical display.

Menu Bar	There are four "pull-down" menus on the Menu Bar of the RCU screen:	
	Control	Allows the RCU Operator to configure the audible alarm, clear the aircraft track history or exit the RCU application. By default, whenever the RCU software is started, the audible alarm will be ON and the track recording will be OFF. Under normal operating conditions, these settings should not be changed. The ability to automatically switch Racks is controlled from this menu item.
	View	Allows the RCU Operator to configure the graphics display.
	Options	Allows the RCU Operator to configure various settings via the Settings and to enter association data that associates an aircraft tail number with their transponder code.
	Help	Allows the RCU Operator to view the RCUx version.

5.4.2.1 Response to Alerts and Alarms

- The RCU Operator should be aware of the following conditions and courses of action.
- Alerts and alarms may be indicated as a result of either BIT or Track monitors.
 - ✓ Alerts: These indicate that one or more key parameters have exceeded an alert limit and may be nearing an alarm limit. Continue normal use of the TLS. If the diagnostic alerts are persistent (indicated continuously for longer than several hours), the RCU Operator should notify the TLS maintenance technician who will determine if a maintenance action is warranted.
 - ✓ Alarm: These indicate that one or more parameters have exceeded a monitor alarm limit. Either the surveillance and/or guidance function is not available for use. If, after the next scheduled BIT, the TLS continues to remain in MAINTENANCE mode, the RCU Operator should call the TLS technician who will coordinate maintenance action.
- Track Alarm: There are a number of reasons why a Track Alarm would occur, and does not mean that the TLS is out of operation. When a track alarm occurs, the current approach will be terminated and the pilot will receive flags indicating no guidance. The TLS Operator will receive a message indicating that guidance was stopped. Continue normal use of the TLS if it remains in Track Mode. If the track alarms are persistent (occur on every approach - or with more than one aircraft) - notify the TLS technician who will determine if a maintenance action is warranted.

5.4.3 RCU Security

The RCU computer shall be used for the express purpose of operating the TLS. No other uses are authorized. To this end, the RCU has been configured to minimize the functions the computer can perform.

Both logon and screen-saver password protections are used on the RCU.

The site operator is responsible for designating authorized RCU Operators, providing RCU Operator security training, and providing required passwords to authorized personnel.

5.5 Rack Switch Controller (optional)

The ability to switch between the main and reserve racks (A & B) is accessed by selecting Switch Racks in the RCU's Control menu [4-27].

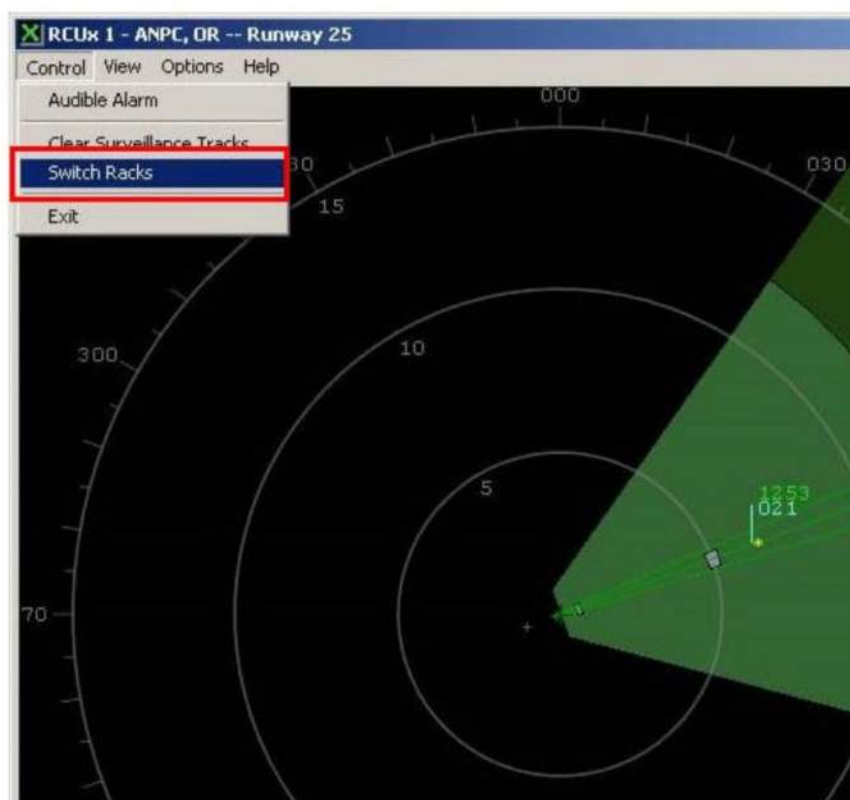


Figure 5-24 RSC Software Interface

5.6 Stand-alone Analysis and Utility Software

Table 5-2 Utility Software Tools

Software Tool Name	TLS File Types	Purpose
Approach	<i>approach.dat</i>	Use to generate approaches and write the approach data to <i>approach.dat</i> .
FlightCheck	<i>N/A</i>	Use to adjust localizer and glide slope alignment and width during flight inspection.
FTM	<i>*.dif</i>	Records TLS and Theodolite data into <i>.dif</i> files, which are used with <i>.raw</i> files for calibration.
GtuTxCal	<i>settings.dat</i>	Use to perform a GTU transmit calibration.
Look	<i>*.raw</i> (This file type is a processed data file)	Displays plots of transponder and Cal/BIT measurements recorded in RAW files during track mode. Monitor Limits can also be displayed for many of these measurements.
MultiBIT	<i>esa.bit, asa.bit, ata.bit, uplink_*.bit, interrog.bit</i> <i>a*b*.ram</i> and <i>e*b*.ram</i>	Displays plots of single/multiple AOA BIT or BIT RAM file measurements. Use for establishing integrity monitor limits and inspecting system health.

RAM	<i>a*.ram, a*.ram, a*b*.ram and e*b*.ram</i> <i>(These file types are raw unprocessed data files)</i>	Allows for the viewing of digitized raw data of transponder replies, Cal/BIT pulses, Interrogation pulses, etc. that are received by the AOA sensors during BIT, surveillance and guidance (RAM recording during BIT and/or Track is required to be enabled in system settings for these "*.ram" files to exist). Note that these files (*.ram) can be extremely large and should only be enabled when specifically needed.
SendToBases	<i>*.dat</i>	Used to select and send updated "*.dat" files to system CPUs.
Survey	<i>survey.dat</i>	Displays a graphical layout of LRUs from survey data.
SyncClock	<i>N/A</i>	Used to set all base CPU clocks to the same date and time as the CPU the tool is running on. Note that the CPU1 and CPU2 clocks are set to the MIU clock when the system software is started from the MI.
SysCal	<i>*.raw and *.dif</i>	Used to read in Calibration flight files and generate "cal.dat" calibration files.
TrackReplay	<i>N/A</i>	Used to replay RCU recorded approach and surveillance data.

5.6.1 Approach Tool

The Approach tool [Figure 5-25] is used to generate or update an approach.dat file. It consists of a plot window and several pull down menus and toolbars. The tool allows design and display of the instrument approach procedure waypoints.

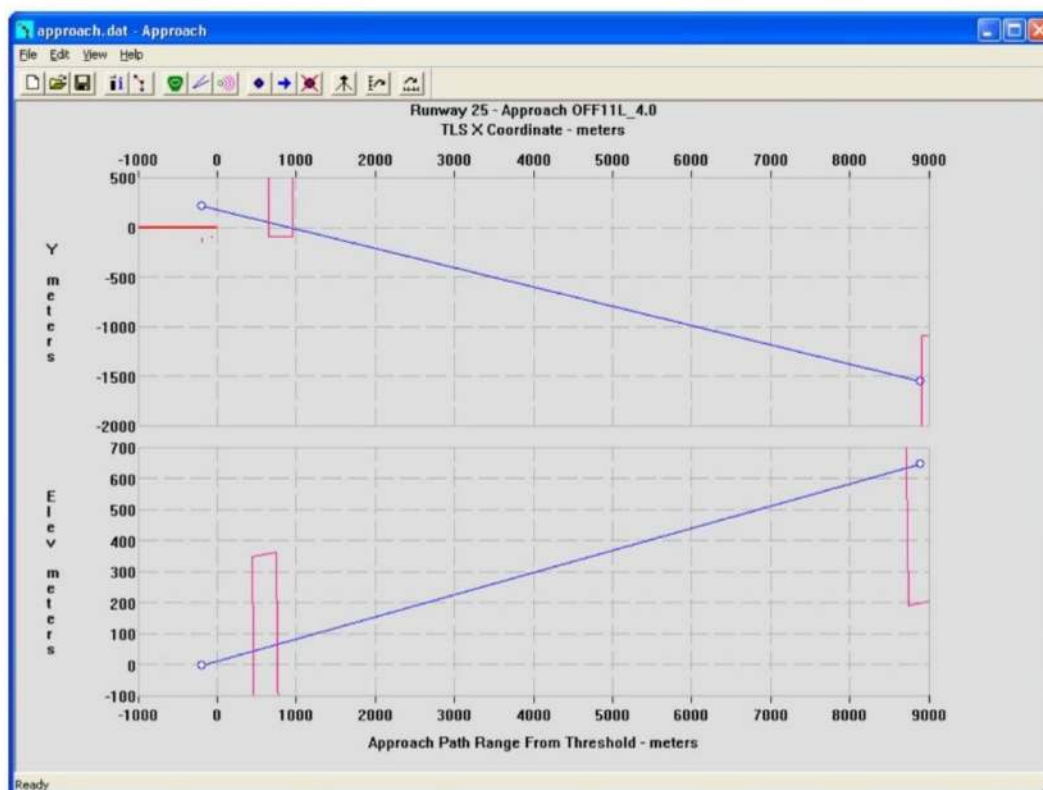


Figure 5-25 Approach Tool Graphical Interface

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5.6.2 Flight Check Tool

Use the FlightCheck tool [Figure 5-26] to adjust TLS localizer and glide slope parameters during flight inspection. The tool is used to:

- Set the course width of the localizer and the path width of the glide slope by writing new parameters to the approach.dat file.
- Set the localizer and glide path alignment by writing new parameters to the cal.dat file.
- Adjust the uplink signal monitor limits for the service volume check at reduced power level by writing parameters to monitor_limits.dat.
- Change the glide slope and localizer ranges and enable recording of track RAM files.
- Start and stop system software to allow changes to take effect.

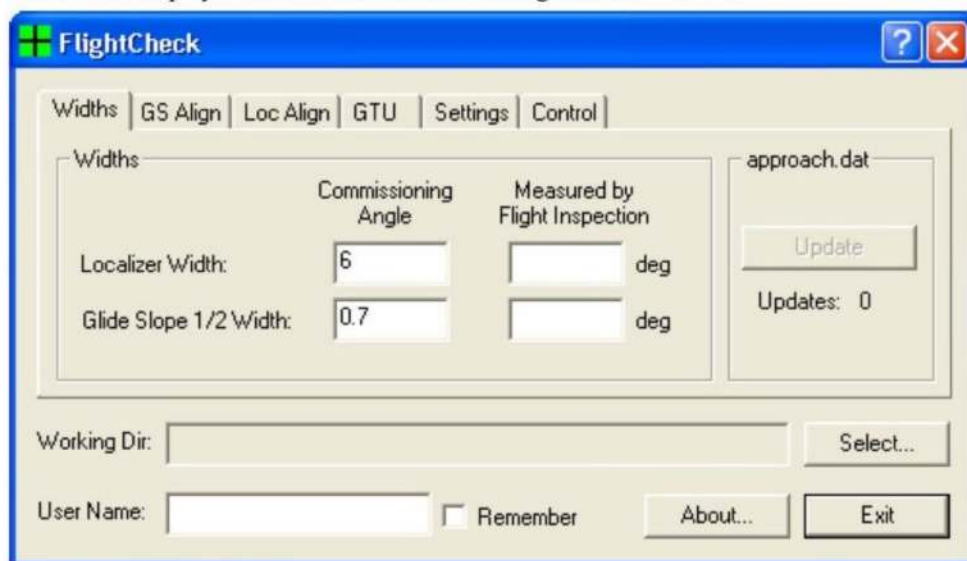


Figure 5-26 Flight Check Tool

5.6.3 Look Tool

The Look tool [Figure 5-27] below is used to graphically display data contained in a TLS .raw file. Raw files are generated by the TLS base software and contain processed sensor data collected for a guidance track. This data is the output of the TLS Pulse Processing software and the input to the TLS Track Processing software, and includes transponder and Cal/BIT measurements (such as TOA, amplitude, phase and frequency) and tilt sensor data.

The tool consists of a plot window and several pull-down menus and toolbars. The pull-down menus and toolbars are used to control the display of raw file data in the plot window. In addition, any part of the plots may be zoomed in or out vertically and/or horizontally through mouse and keyboard controls. Data can be displayed for the ESA, ASA or both. The monitor limits may be displayed if the system monitor_limits.dat file is in the directory of the raw file selected or in its parent directory.

Refer to the Look tool help menu and the TLS software tools supplement for more information.

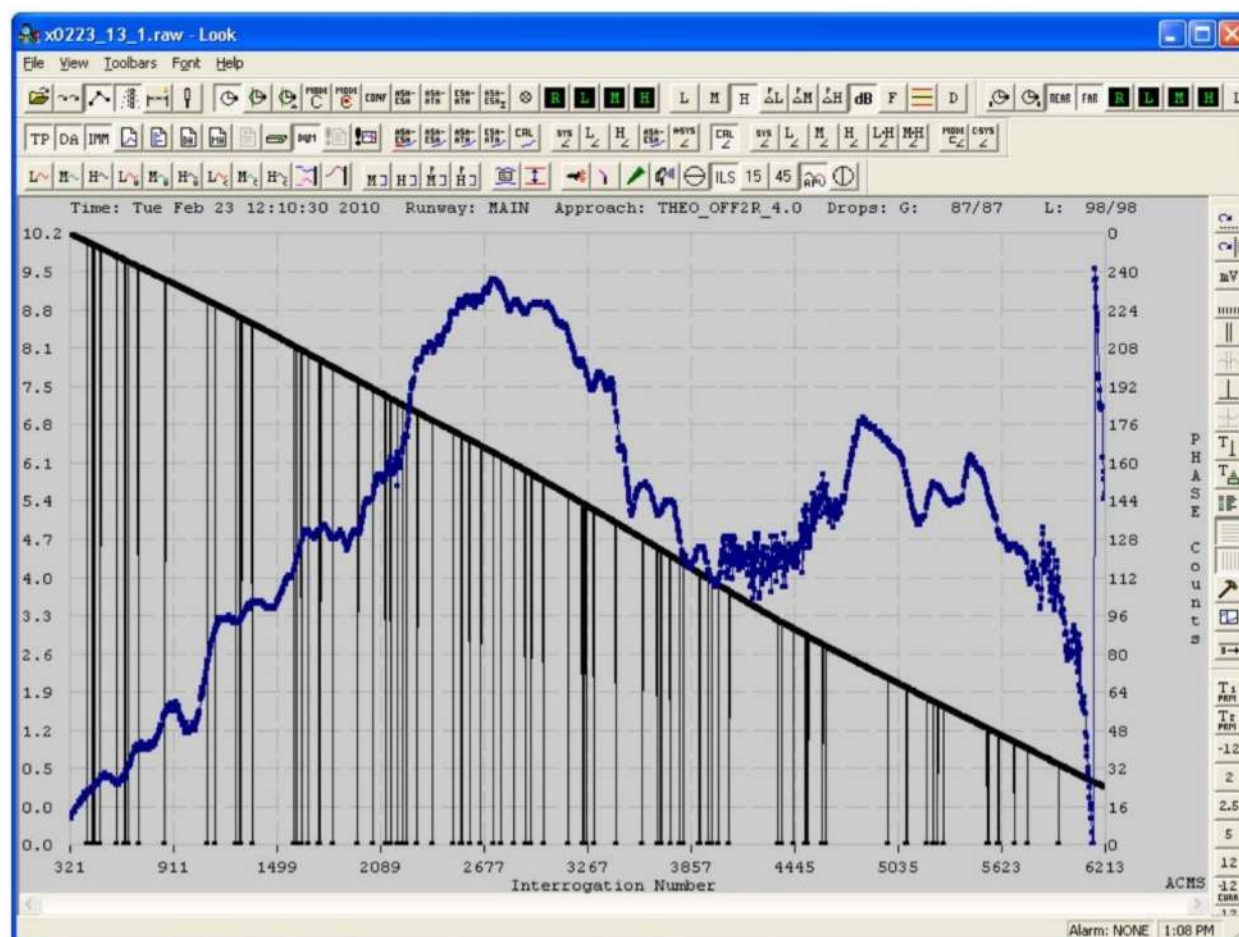


Figure 5-27 Look Tool

5.6.4 MultiBIT Tool

The MultiBIT tool [Figure 5-28] is used to graphically display data contained in TLS AOA BIT files (ESA.bit, ASA.bit, ATA.bit, uplink_*.bit and interrog.bit) or BIT RAM files (a*b*.ram and e*b*.ram). These files are generated by the TLS base software and contain sensor health data collected during BIT. This data includes TOA, amplitude, phase, temperature, frequency and tilt (ESA only) measurements.

The tool consists of a plot window and several pull-down menus and toolbars. The pull-down menus and toolbars are used to control the display of data in the plot window. In addition, any part of the plots may be zoomed in or out vertically and/or horizontally through mouse and keyboard controls. The monitor limits may be displayed if the system monitor_limits.dat file is in the directory where BIT or RAM files are selected or in its parent directory.

Refer to the MultiBIT tool help menu and the TLS software tools supplement for more information.

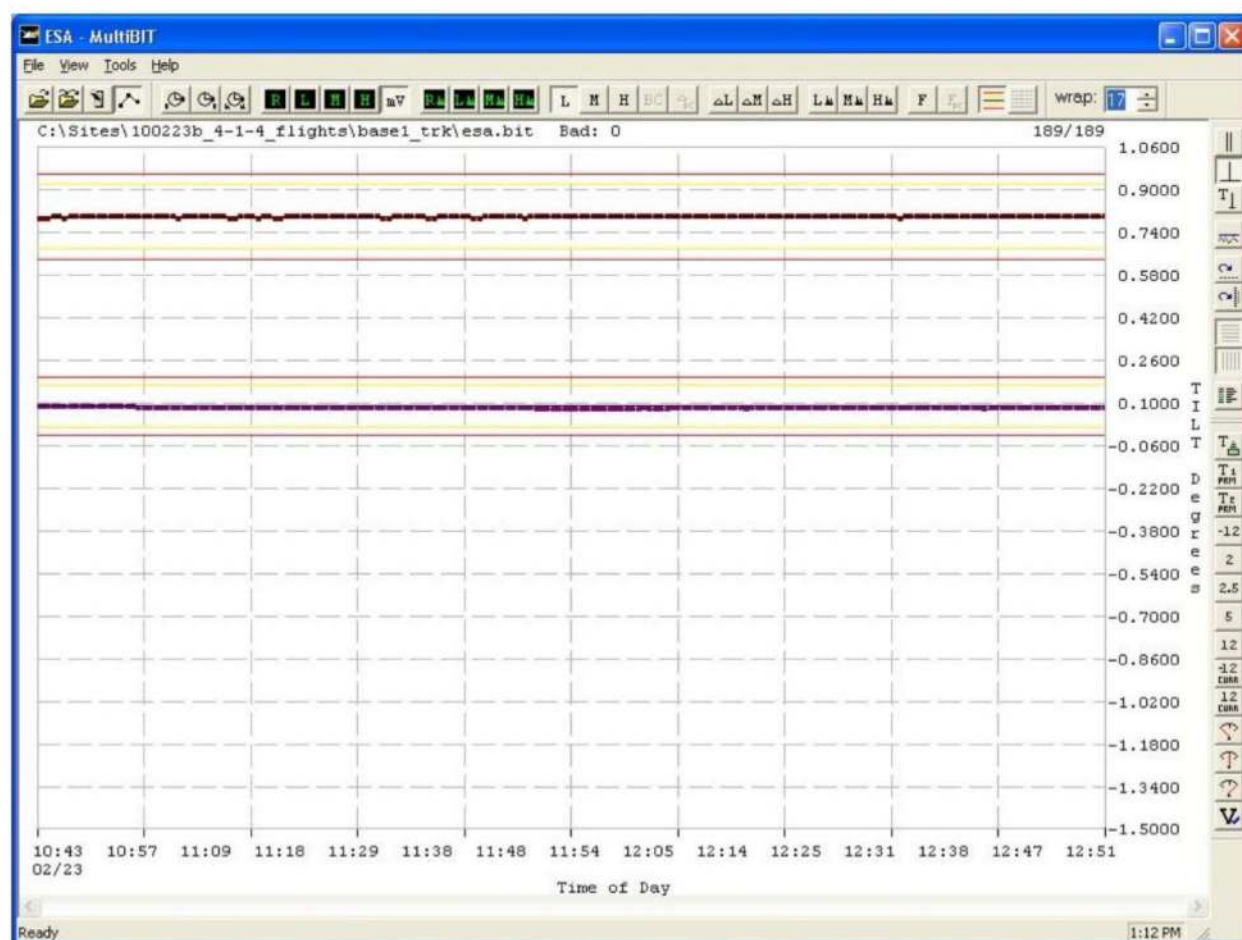


Figure 5-28 MultiBIT Tool

5.6.5 RAM Tool

The RAM tool [Figure 5-29] is used to graphically display data contained in a TLS .ram file (or RAM file).

RAM files are generated by the TLS base software and contain raw sensor data collected during BIT or in Test and Track Modes. Sensor data includes amplitude, phase and frequency measurements.

The tool consists of a plot window and several pull-down menus and toolbars. The pull-down menus and toolbars are used to control the display of sensor data in the plot window. In addition, any part of the plots may be zoomed in or out vertically and/or horizontally through mouse and keyboard controls. The primary function of the RAM tool is to plot sensor amplitudes. These plots provide the user with a means to perform troubleshooting tasks, such as determining if the Cal/BIT pulse is present. The tool also provides transponder pulse and bracket characteristics as a result of running the TLS Pulse Processing software.

Refer to the Ram tool help menu and the TLS software tools supplement for more information.

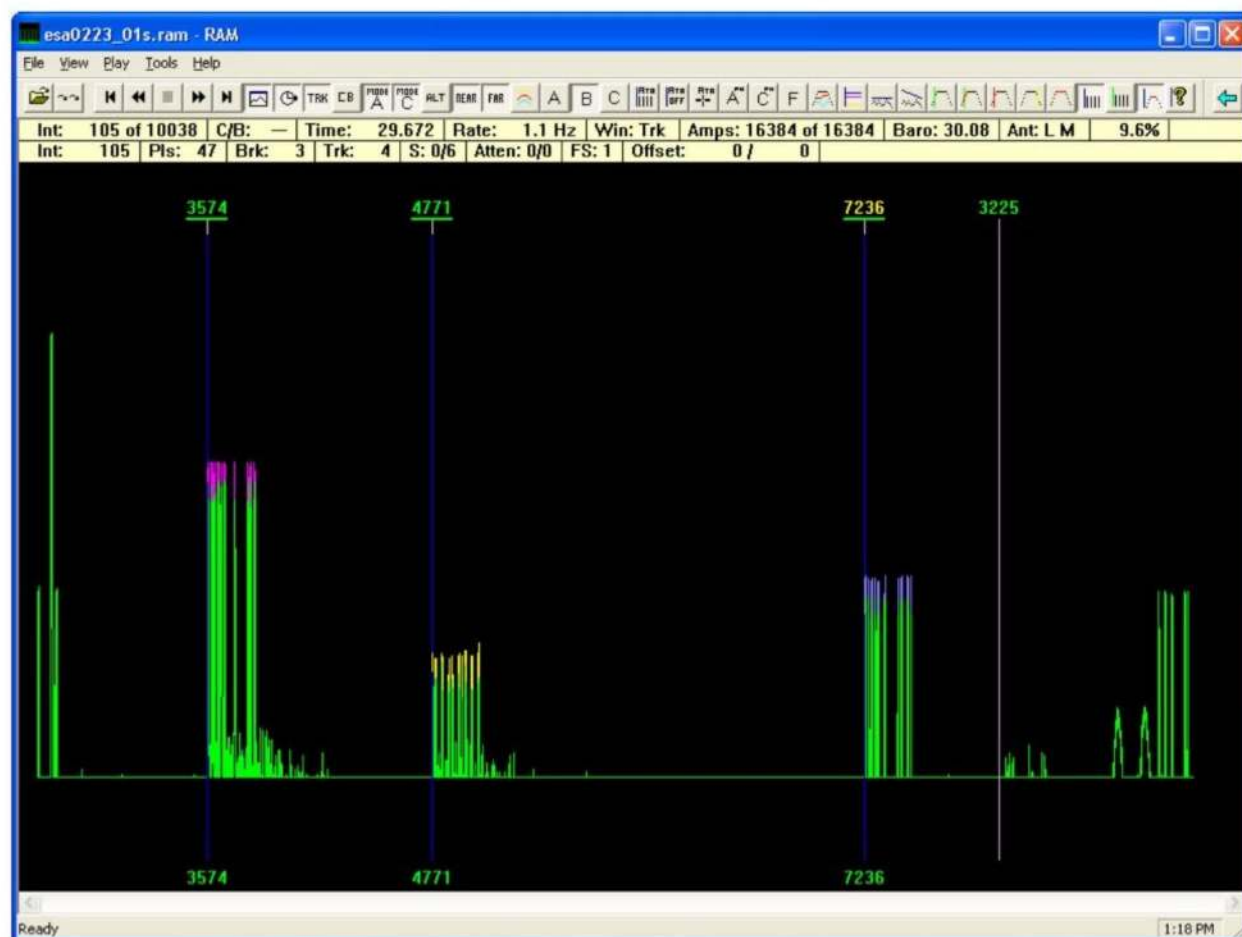


Figure 5-29 RAM Tool

5.6.6 SendToBases Tool

The SendToBases tool [Figure 5-30] is used to distribute TLS configuration data files to the Base1 and Base2 computers. Files are sent to Base1 using FTP and to Base2 using a network copy. The tool consists of a single dialog that allows input of the source file location and the destination computers. When a send is performed, the tool displays the results in a field at the bottom of the window.

The user must stop the system software and exit the MI application before sending files to Base1 or Base2 using this tool. Once the files have been distributed, the software on each computer will read the new data files when the software is restarted.

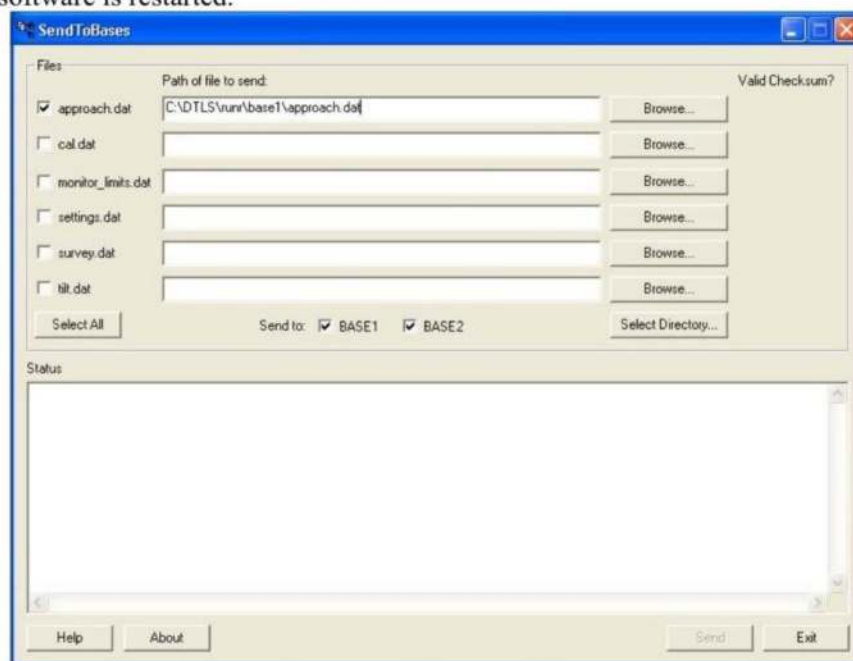


Figure 5-30 SendToBases Tool

5.6.7 SyncClock Tool

The SyncClock tool [Figure 5-31] is used to synchronize the clocks on the base computers. The tool consists of a single dialog that displays the date and time on the local computer. Pressing the **Sync** button causes the date and time on the Base1 and Base2 computers to be set to the date and time on the local computer (the hardware clocks are also set). Pressing the **Close** button exits the tool.

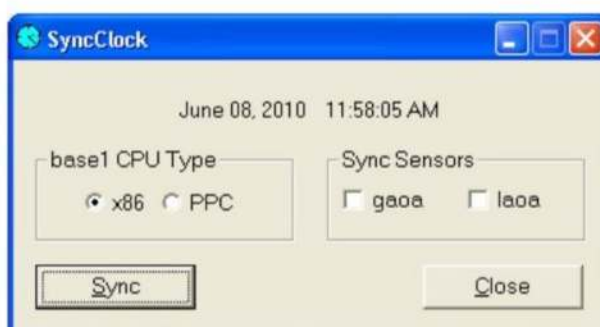


Figure 5-31 SyncClock Tool

5.6.8 SysCal Tool

The SysCal tool [Figure 5-32] is used to generate or update a calibration configuration data file cal.dat. The tool consists of a plot window and several pull-down menus and toolbars. The pull-down menus and toolbars are used to set processing options and control the display of data in the plot window.

The tool provides the capability to generate the calibration file from the following sources:

- Site data.
- Raw file with theodolite data.
- Raw file without theodolite data.

The tool also provides the capability to generate and display a replay. A replay shows the azimuth and elevation tracks and the difference between them and the theodolite data.

The purpose of calibration is to align the measurements with the runway. The principle part of calibration and the part that requires the most attention is alignment of the ESA phase measurements. This consists of using the SysCal tool to fit a curve to data points as shown in Figure 5-32.

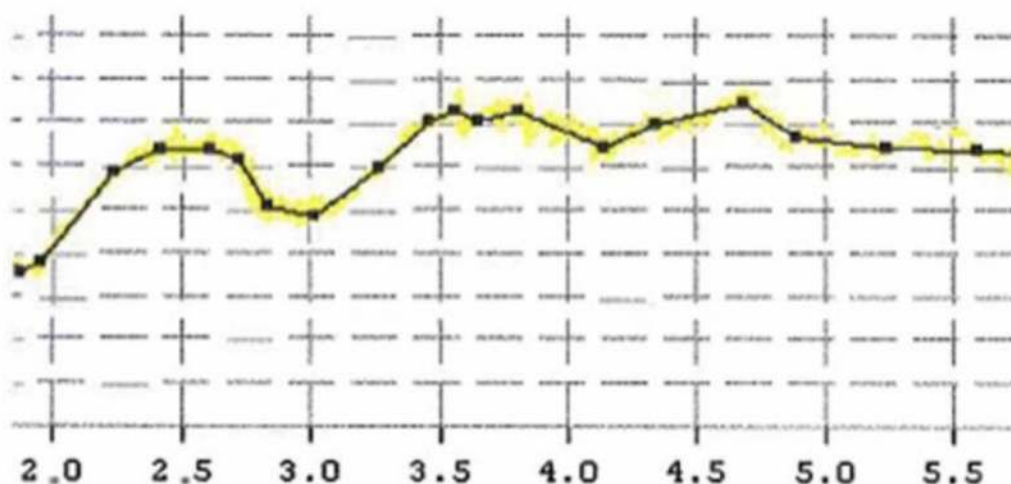


Figure 5-32 Sample Calibration Curve for ESA

A sample portion of a calibration curve is shown in Figure 5-32. Each individual dot (yellow) represents a phase data measurement of the aircraft position. Multipath returns from the ground cause a non-linear relationship between the measured angle and the actual angle. Note that the “calibration curve” (represented by the square dots connected by lines) follows the average

5.6.9 TrackReplay Tool

The TrackReplay tool [Figure 5-33] is used to replay RCU recorded approach and surveillance data. It consists of a plot window (similar to the RCUX application), replay controls, and pull-down menus. The **Browse...** button is used to select the directory containing recorded data (e.g., C:\run\trk on the RCU).

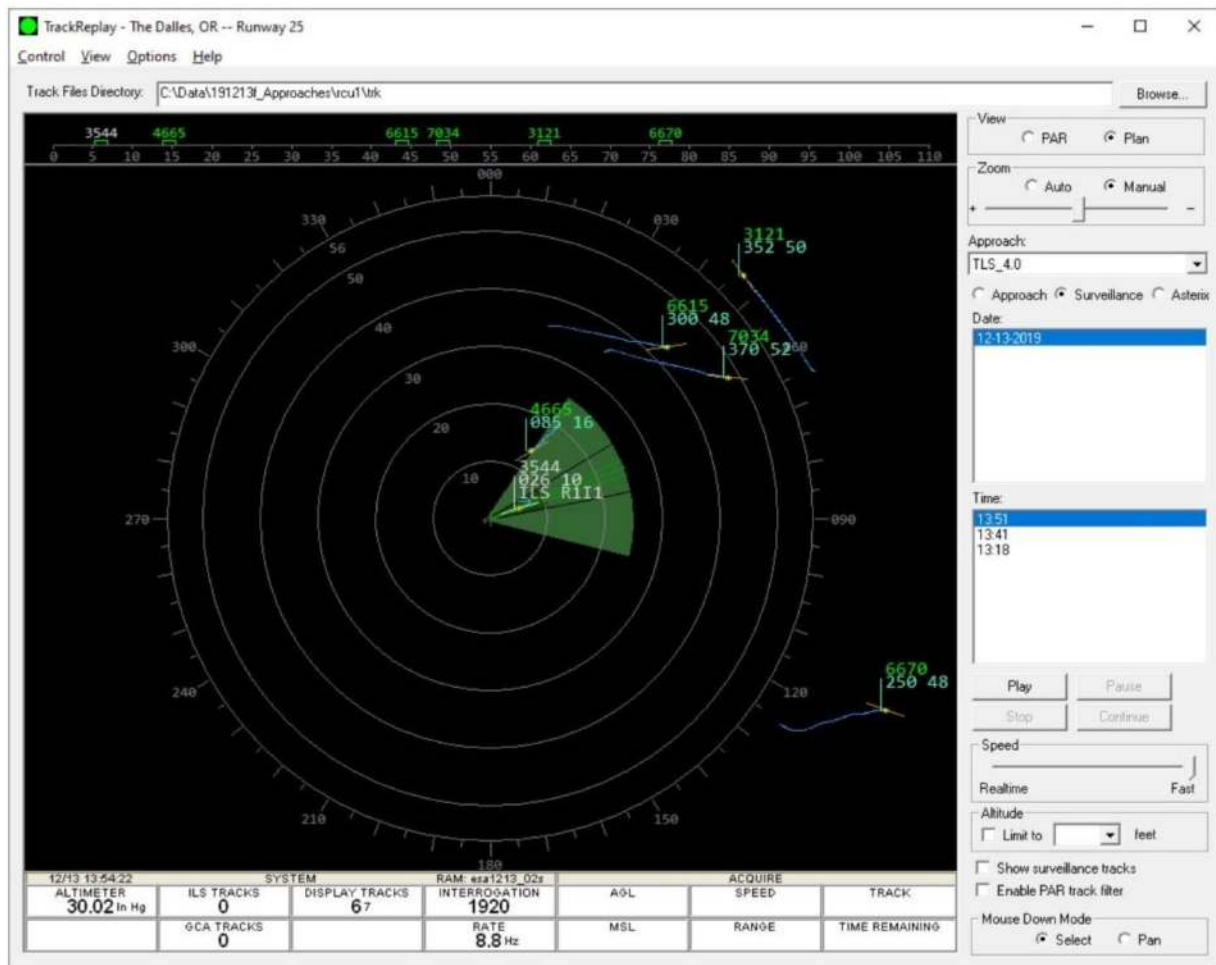


Figure 5-333 TrackReplay Tool

Appendix A Abbreviations and Definitions

Abbreviation	Definition
ANPC	Advanced Navigation & Positioning Corporation
ASA	Azimuth Sensor Assembly (previously called LAOA)
ATA	Azimuth Time-Of-Arrival (TOA) Antenna
ATC	Air Traffic Control
BIT	Built-In-Test
Cal/BIT	Calibration/Built-in-Test
CDI	Course Deviation Indicator
CNTL	Control
CPU	Central Processing Unit
DDM	Differential Depth of Modulation
DME	Distance Measurement Equipment
ESA	Elevation Sensor Assembly (previously called GAOA)
FTM	Flight Test Monitor
GAOA	Glide slope Angle-of-Arrival (previous term for ESA)
GS	Glide slope
GTU	Guidance Transmitter Unit
HSI	Horizontal Situation Indicator
Hz	Hertz
ICC	Interrogator Control Card
ID	Identification
IF	Intermediate Frequency
IFR	Instrument Flight Rules
ILS	Instrument Landing System
IM	Integrity Monitor
KVM	Keyboard Video Mouse
LAOA	Localizer Angle-of-Arrival (previous term for ASA)
LOC	Localizer
LRU	Line Replaceable Unit
nm	nanometer
NM	Nautical Mile
MHz	Mega Hertz
MI	Maintenance Interface
MIU	Maintenance Interface Unit
OS	Operating System
PRC	Phase Receiver Card
RCU	Remote Control Unit
RF	Radio Frequency

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A-1

Abbreviation	Definition
RSA	Runway Safety Area
RSC	Reserve Rack and Rack Switch Control
RSM	Remote Status Monitor (optional feature)
SCI	Software Configuration Item
SSR	Secondary Surveillance Radar
SDM	Sum of Depth of Modulation
TLS	Transponder Landing System
UHF	Ultra High Frequency
UPS	Uninterruptible Power Supply
VAC	Volts Alternating Current
VFR	Visual Flight Rules
VHF	Very High Frequency

Appendix B ANPC TLS Manual Recommended Change Form

TLS TECHNICAL INSTRUCTION MANUAL RECOMMENDED CHANGE FORM

The following form is intended to document recommended changes pertaining to the Technical Instruction Manual. The TLS TI Manual Recommended Change Form may be legibly handwritten or typed and must include the revision level of the TI Manual being discussed. The recommendation sheet is intended to be used as master copies from which to make working recommendation sheet.

Scan and email the completed form to Email CustomerSupport@anpc.com or use standard mail to:

ANPC
489 N. 8th Street, Suite 203
Hood River, OR 97031
ATTN: Customer Support

Phone (541) 386-1747

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B-1

TLS MANUAL RECOMMENDED CHANGE FORM

This recommended change pertains to:

- ☐ Safety
- ☐ Operation
- ☐ Maintenance
- ☐ Documentation

Recommendation:

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

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